

Spectral Sequences of Flow Categories

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Abstract

Framed flow categories and their stable homotopy types are central tools in modern Floer, contact, Khovanov, and knot homotopy theories. In this paper, given an $\mathcal{X}$ -structured flow category $\mathcal{C}$ and a generalized homology theory A equipped with a map $M\mathcal{X} \rightarrow A$, we construct an Atiyah-Hirzebruch-type spectral sequence which converges to the homology of the realization $A_*(|\mathcal{C}|)$. Moreover, if A is a bordism theory - for example, if $\mathcal{C}$ is framed and A is stable homotopy - we provide a description of the higher pages and differentials in terms of gluing the flow manifolds in $\mathcal{C}$ to a geometric analog of Toda brackets, an idea traced back to Floer[13].

This work provides a bordism-level perspective on spectral sequences, aligning with the approach of Abouzaid and Blumberg, who developed a bordism model for spectra. We demonstrate these spectral sequence techniques by calculating the unoriented and oriented bordism homology of real projective spaces using Morse flow data.

Independent of flow categories, we provide a description of the spectral sequence of a chain complex in a stable category in terms of morphisms between chain complexes.

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1 Introduction

****Note for rewriting.**** In the first section after the introduction, motivate the Toda-manifold construction before its full definition. A small d_2 example should compare the two algebraic nullhomotopies with the two manifold pieces glued along their common flow boundary, so the reader sees why this particular gluing represents the differential.

We change the intro to use "flow bimod", now we need to shchange the rest of the paper

Short Introduction

Flow categories originally emerged as a categorical bookkeeping mechanism to organize the geometric and analytic data extracted from Hamiltonian systems on a manifold [9]. In recent years, flow categories and their stable homotopy types have become central tools for lifting invariants in Floer, contact, and Khovanov homology into the realm of stable homotopy theory. A natural question arises: how can we systematically compute the generalized homology of these realization spectra directly from the underlying geometric flow data?

Here is an intuitive description of the main theorem of the paper using a toy model coming from Morse theory. Let M be a closed manifold and $f : M \rightarrow \mathbb{R}$ a Smale Morse function. Denote by $\text{Crit}(f)$ the set of critical points of f , and by $\text{ind} : \text{Crit}(f) \rightarrow \mathbb{Z}$ the index function. For simplicity, assume that there is one critical point for each index $i \in \{0, \dots, n\}$, and denote the critical points by x_i where $\text{ind}(x_i) = i$. The first result is:

There is a spectral sequence

$$E_1^{n,p} = \Omega_{n-p}^{\text{fr}} \implies \pi_n \Sigma_+^\infty M.$$

where Ω_*^{fr} is the framed bordism ring, and $\Sigma_+^\infty M$ is the stable homotopy type of M .

Let x_j, x_i be two critical points of f with $j > i$. We denote by $\mathcal{M}(x_j, x_i)$ the compactification of the moduli space of flow lines from x_j to x_i , up to reparametrization. Recall that $\mathcal{M}(x_j, x_i)$ is a framed manifold with corners of dimension $\text{ind}(x_j) - \text{ind}(x_i) - 1$, and that the boundary structure is given by

$$\partial \mathcal{M}(x_j, x_i) = \bigcup_{j > r > i} \mathcal{M}(x_j, x_r) \times \mathcal{M}(x_r, x_i).$$

Let W_p be a representative of a bordism class in $E_1^{n,p} = \Omega_{n-p}^{\text{fr}}$. In our framework, we view W_p as a right flow module of degree n over the trivial subcategory containing only the critical point x_p . The main theorem states that W_p survives to the $(r+1)$ -th page (also denoted by $W_p \in Z_{r+1}$) if and only if it ****extends to a right flow module**** W of degree n over the truncated flow category $\mathcal{C}_{[p, p-r]}$. Concretely, this means providing a family of framed manifolds $\{W_s\}$ for $s \in \{p, \dots, p-r\}$ of dimension $n - s$ such that the corner structure maps

$$W_t \times \mathcal{M}(x_t, x_s) \hookrightarrow W_s$$

satisfy the axioms of a right flow module.

To compute the differential, we glue this right module W together with the canonical left flow module $\mathcal{M}(-, x_{p-(r+1)})$. Let $t, u \in \mathbb{Z}$ be such that $p \geq t > u \geq p - (r+1)$. By the boundary structure axioms of the right module W and the left module $\mathcal{M}(-, x_{p-(r+1)})$, we have the following two embeddings:

$$W_t \times \mathcal{M}(x_t, x_{p-(r+1)}) \hookleftarrow W_t \times \mathcal{M}(x_t, x_u) \times \mathcal{M}(x_u, x_{p-(r+1)}) \hookrightarrow W_u \times \mathcal{M}(x_u, x_{p-(r+1)})$$

and these boundary components are shared only by these two manifolds of the collection $W_s \times \mathcal{M}(x_s, x_{p-(r+1)})$. Because all boundary components of $W_s \times \mathcal{M}(x_s, x_{p-(r+1)})$ are of this form, we can glue the collection together along these shared faces to obtain a framed closed manifold. We denote this closed manifold by $W \circ \mathcal{M}(-, x_{p-(r+1)})$, and call it a Toda manifold:

$$W \circ \mathcal{M}(-, x_{p-(r+1)}) := \bigcup_{s \in \{p, \dots, p-r\}} W_s \times \mathcal{M}(x_s, x_{p-(r+1)}).$$

This gluing operation is a geometric realization of the composition of modules (sometimes called a geometric tensor product).

To verify that this manifold represents a class in the correct group for the differential, we compute its dimension. For each s , the pieces being glued have dimension:

$$\dim(W_s \times \mathcal{M}(x_s, x_{p-(r+1)})) = \dim(W_s) + \dim(\mathcal{M}(x_s, x_{p-(r+1)})) = (n-s) + (s - (p-(r+1)) - 1) = n - p + r.$$

Note that the -1 in the second term arises from the dimension formula for flow manifolds. The result $n - p + r$ matches the expected dimension for an element of the $(r+1)$ -th page. The last part of the theorem states that if W_p survives to Z_{r+1} , then the $r+1$ differential is given by the bordism class of this Toda manifold:

$$d_{r+1}(W_p) = [W \circ \mathcal{M}(-, x_{p-(r+1)})] \in \Omega_{n-p+r}^{\text{fr}} = E_{r+1}^{n-1, p-(r+1)}.$$

****Note for rewriting.**** The closed manifold gives a framed-bordism class on the E_1 page. Its class on the $(r+1)$ -st page is the differential; the full bordism group is generally not equal to that later-page group. State how the page class is obtained and account for the choice of extension W . We need to move from E_r to $\mathcal{E}_r$ or to write E_1

See Fig. 2 for an example of the structure of the right module W and the flow manifolds, Fig. 3 for the gluing diagram of the Toda manifold, and Fig. 4 for the resulting closed manifold.

Background

Morse theory basics. The field of differential topology studies the connection between analysis on a smooth n -manifold M and its underlying topology. This field was pioneered by Morse in the beginning of the 20th century, and further developed by Smale, Thom, Milnor and more (see [24] for Morse first work and [22] for Milnor’s classic exposition of the subject). To begin understanding M , we take a “nice” (namely a Smale-Morse) function $f: M \rightarrow \mathbb{R}$. We extract the *critical points* of f and, for any critical point p , assign its *index* $\text{ind}(p)$ - which is defined as the signature of the Hessian of $-\nabla f$. Pictorially, this index represents the number of ‘downward’ axes at p , see Fig. 1.

For any two critical points y and x , we define $M(y, x)$ to be the moduli space of curves (up to reparametrization) in M that start at y and end at x , and follow the gradient field $-\nabla f$. We call such curves *flow lines*; see Fig. 1. Under generic conditions, $M(y, x)$ is a smooth (not necessarily closed) manifold of dimension $\text{ind}(y) - \text{ind}(x) - 1$.

Framed flow manifolds. Our main interests are *flow manifolds*, which are the compactification

$$\mathcal{M}(y, x) := \overline{M(y, x)}.$$

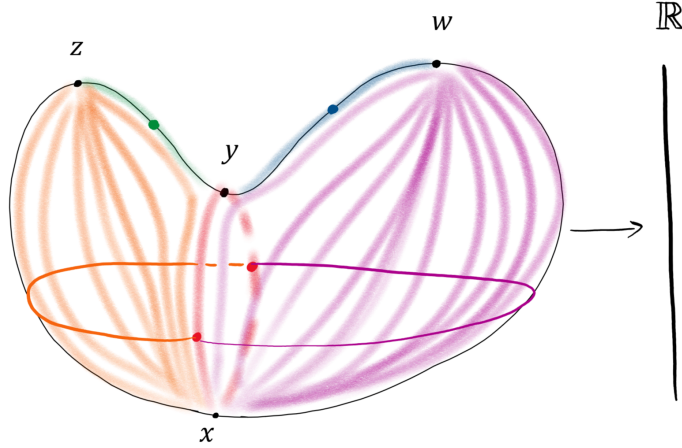


Figure 1: A Smale-Morse function on the sphere, given as projection to the z axis of the heart-shaped embedding. The letters represent critical points (with indices 2,2,1,0), the flow lines are drawn as soft, colorful lines, the solid colorful dots are representatives for the zero-dimensional flow manifolds and the solid colorful lines are representatives for the one-dimensional flow manifolds (notice the boundary structure as broken flow lines)

The compactification is given by adding *broken flow lines*, which are piecewise smooth curves that start at y and end at x and pass through other critical points. We get that the flow manifolds have a *corner structure*

$$\partial\mathcal{M}(z, x) = \bigcup_{\text{ind}(z) > \text{ind}(y) > \text{ind}(x)} \mathcal{M}(z, y) \times \mathcal{M}(y, x).$$

see Fig. 1. The fact that $\mathcal{M}(y, x)$ is constructed from the transverse intersection of two discs (one for curves ascending to y and one for curves descending to x) gives a stable *framing* on $\mathcal{M}(y, x)$ - a trivialization of the form

$$T\mathcal{M}(y, x) \oplus \mathbb{R}^{\ell_1} \simeq \mathbb{R}^{\ell_2}$$

where $\mathbb{R}^{\ell_i}$ are trivial vector bundles over $\mathcal{M}(y, x)$. Moreover, the framing of all the flow manifolds is compatible with the corners structure.

Morse homology. It's worth saying that this data is sufficient to calculate the homology $H_*(M, \mathbb{Z})$. It is given as the homology of the Morse chain complex (C_*, d) . The generators of C_p are critical points with index p . Given two critical points y, x with $\text{ind}(y) - 1 = \text{ind}(x)$, the flow manifold $\mathcal{M}(y, x)$ is in fact compact; the coefficient of x in $d(y)$ is given by the signed count of $\mathcal{M}(y, x)$, where the signs are determined by the framing. The boundary structure of the one-dimensional moduli spaces $\mathcal{M}(z, x)$ shows that $d^2 = 0$. See [28].

Bordism and generalized homology. The fact that the flow manifolds are framed hints at an important theorem in differential topology, the Pontryagin-Thom isomorphism. We say that two closed framed n -manifolds M_1 and M_2 are *bordant* if there is a framed $(n + 1)$ -manifold W , with $\partial W = M_1 \sqcup M_2$, in a way that is compatible with the framing. We denote by Ω_n^{fr} the set of n dimensional framed manifolds up to bordism. In fact the collection Ω_*^{fr} form a graded ring, by disjoint union and Cartesian product.

The *Pontryagin-Thom isomorphism* states that there is an isomorphism

$$\Omega_n^{\text{fr}} \simeq \operatorname{colim}_k [S^{n+k}, S^k]$$

between bordism groups of framed manifolds and stable homotopy groups of spheres (see [25, 30]). The right hand side is also known as the coefficient ring $\pi_n(\mathbb{S})$, where $\mathbb{S}$ is the spectrum representing stable homotopy. We can view the Morse chain complex as a chain complex with coefficient ring $\mathbb{Z} = \Omega_0^{\text{fr}} = \mathbb{S}_{\leq 0}$ and the differential is given by multiplication by the 0 dimensional flow manifolds $\mathcal{M}(y, x)$.

Floer theory A related question is the analysis of real-valued functions on objects that are not finite-dimensional closed manifolds. Floer developed this idea in the 1980s, leading to what is now known as Floer theory. In general, Floer’s idea is to generate a chain complex in which the differentials are defined by ‘counting solutions’ to a moduli problem (see [12] for initial formulations of this idea). In this case, the outcome does not yield the homology of a manifold, and one may ask: *Are Floer homology groups merely the shadow of a more structured object?*

Flow categories. The development of a framework to address these questions led Cohen, Jones, and Segal (CJS) in [9] to introduce *framed flow categories*, which axiomatize the data extracted from a Smale-Morse function on a manifold. In short, a framed flow category $\mathcal{C}$ consists of a set of objects (corresponding to critical points) with a grading function $|\cdot|$ (corresponding to the index). For two objects $y, x \in \mathcal{C}$, the morphism space from y to x is an $(|y| - |x| - 1)$ -dimensional manifold with corners, denoted $\mathcal{M}_{y,x}$, equipped with a choice of trivialization of $T\mathcal{M}_{y,x} \oplus \mathbb{R}^k$ for some k (corresponding to the stable framing). Moreover, the composition is given by identifying the faces in the following way:

$$\mathcal{M}_{z,y} \times \mathcal{M}_{y,x} \hookrightarrow \mathcal{M}_{z,x}.$$

See Section 3.3 for definitions and details. Another building block in the theory are *flow modules*. Given a flow category $\mathcal{C}$, a *right flow module* W over $\mathcal{C}$ (see Definition 3.24 for the formal definition) consists of a collection of manifolds $\{W_x\}_{x \in \mathcal{C}}$, together with corner structure maps

$$W_y \times \mathcal{M}_{y,x} \hookrightarrow W_x.$$

A right flow module is of *degree i* if

$$\dim W_x = i - |x|$$

for each pair of objects y and x in $\mathcal{C}$. See [26] definition 4.6, 4.7. By abuse of notation, we denote the collection of right modules of degree i over $\mathcal{C}$

$$[*[i], \mathcal{C}].$$

Similarly, we can define left flow modules of degree i over $\mathcal{C}$ (see Definition 3.25 for the formal definition) to be collections of manifolds $\{V_x\}_{x \in \mathcal{C}}$, together with corner structure maps

$$\mathcal{M}_{y,x} \times V_x \hookrightarrow V_y.$$

such that $\dim V_x = |x| - i$. We denote the collection of left modules of degree i over $\mathcal{C}$ by

$$[\mathcal{C}, *[i]].$$

A useful example for left module on a category $\mathcal{C}$ is given by fixing $x' \in \mathcal{C}$ and taking the collection

$$\{\mathcal{M}_{x,x'}\}_{x \in \mathcal{C}}. \quad (1)$$

Indeed we have that the corner structure is given by the maps

$$\mathcal{M}_{y,x} \times \mathcal{M}_{x,x'} \hookrightarrow \mathcal{M}_{y,x'}$$

The degree of this module is given by

$$|x| - \dim(V_x) = |x| - (|x| - |x'| - 1) = |x'| + 1$$

and we denote it simply by $\mathcal{M}_{-,x'}$.

Filtrations and coherent chain complexes. Another result of CJS was to attach a stable homotopy type to a framed flow category by the following idea.

Given a filtered space $X_0 \rightarrow X_1 \rightarrow \dots \rightarrow X_n = X$ we can take the successive cofibers to produce a “chain complex” defined by

$$K_* := X_n/X_{n-1} \xrightarrow{d} \Sigma(X_{n-1}/X_{n-2}) \xrightarrow{d} \dots \xrightarrow{d} \Sigma^n X_0,$$

where d^2 is null-homotopic via the canonical null homotopy arising from X . Moreover, viewing K_* from higher-categorical perspective allows us to encode additional data. In particular, there exist two null homotopies for d^3 that can be concatenated to form a loop in a suitable mapping space; the chain complex is then equipped with a null homotopy of this loop, and so forth. We call this structure a *coherent chain complex*. We can attempt to recover X by taking *iterated cofibers*. For example, note that

$$\Sigma^2 X_1 = \text{cof}(\Sigma^1 X_1/X_0 \rightarrow \Sigma^2 X_0), \quad \Sigma^2 X_2 = \text{cof}(\text{cof}(X_2/X_1 \rightarrow \Sigma X_1/X_0) \rightarrow \Sigma^2 X_0)$$

and indeed, we can recover X up to a certain suspension. See [Section 2.1](#) and [\[5\]](#) for full detailed text.

Building on this idea, CJS showed that a framed flow category $\mathcal{C}$ can be collapsed, via a Pontryagin-Thom-type construction, into a coherent chain complex K in Spectra. After taking iterated cofiber, we obtain a spectrum

$$|\mathcal{C}| := \lim_p \text{cof}(\text{cof}(\dots \text{cof}(\text{cof}(K_p \rightarrow K_{p-1}) \rightarrow K_{p-2}) \rightarrow \dots \rightarrow K_1) \rightarrow K_0)$$

which is called the *realization* of $\mathcal{C}$.

Remark. Note that $|\mathcal{C}|$ depends on the choice of framing for the manifolds $M_{b,a}$.

Example - Morse flow category. To combine the ideas of framed flow categories and chain complexes, let $f : M \rightarrow \mathbb{R}$ be a manifold equipped with a Smale-Morse function. First, we construct from f a framed flow category $\mathcal{C}_f$ - by taking $\mathcal{M}_{y,x} := \mathcal{M}(y,x)$, then we collapse it to a chain complex $K_{\mathcal{C}_f}$ and realize it as $|\mathcal{C}_f| \in \mathbf{Sp}$. A result by CJS [\[9\]](#), states that this procedure recovers the stable homotopy type of M :

$$|\mathcal{C}_f| \simeq \Sigma_+^\infty M.$$

This framework has proven useful in Floer theory, where the solutions to the moduli problem sometimes exhibit the structure of a flow category $\mathcal{C}$. In such cases, we can realize the flow category, demonstrating that Floer homology is indeed a shadow of $|\mathcal{C}|$.

Looking back to Morse homology, we observe that the homology of a flow category can be determined from its zero-dimensional morphism manifolds. The fact that we can recover the stable homotopy type of M from its framed flow category motivates the idea that we can read off the (generalized) homology of M using the flow manifolds.

This idea goes back to Floer, who pointed out in [13] that: If h_* is a general homology theory, trajectory space analysis suggests contributions from compact components of $\mathcal{M}(x, y)$. For stable homotopy $h_* = \pi_*^s$, this corresponds to the framed cobordism type. While higher-dimensional components of $\mathcal{M}(x, y)$ may not be compact, a modified approach could yield a spectral sequence. This framework has limited use in finite-dimensional Morse theory but may apply to infinite-dimensional settings.

Notations

We denote by **Top** the 1-category of topological spaces, by **Spaces** the category of homotopy types (or ∞ -groupoids), and **Sp** is the category of spectra. The stabilization functor is denoted $\Sigma_+^\infty : \mathbf{Spaces} \rightarrow \mathbf{Sp}$, and $\widehat{\mathbf{Ch}}(\mathbf{Sp})$ and $\widehat{\mathbf{Fil}}(\mathbf{Sp})$ denote the categories of coherent chain complexes in Spectra and (complete) filtered Spectra, respectively.

For a topological space Y , we denote by Y_+ its one-point compactification (if Y is compact, then $Y_+ = Y \cup \{*\}$). For a CW complex Y , we denote by $\mathrm{sk}_n Y$ its n -th skeleton.

For a stable tangential structure $\xi : \mathcal{X} \rightarrow BO$, we denote by $\Omega_*^{\mathcal{X}}$ the graded $\mathcal{X}$ -bordism ring and by $M\mathcal{X}$ the Thom spectrum associated with the $\mathcal{X}$ -structure (see Section 7 for definitions).

In this paper, a manifold is assumed to be a *closed manifold* unless stated otherwise.

Let $\mathcal{C}$ be a flow category. For $q, p \in \mathbb{Z}$ we denote by $\mathcal{C}_{[q,p]}$ the full subcategory of $\mathcal{C}$ consisting of objects satisfying $q \geq |x| \geq p$, and we define $\mathcal{C}_p := \mathcal{C}_{[p,p]}$. For brevity, we write

$$M_{p,q} := \bigsqcup_{y \in \mathcal{C}_q, x \in \mathcal{C}_p} M_{y,x}.$$

for the union of the flow manifolds, and for a right / left module over $\mathcal{C}$ we write

$$W_p := \bigsqcup_{x \in \mathcal{C}_p} W_x, \quad V_p := \bigsqcup_{x \in \mathcal{C}_p} V_x.$$

Moreover, for a stable tangential structure $\xi : \mathcal{X} \rightarrow BO$, one can define $\mathcal{X}$ flow categories as those flow categories in which the morphism manifolds are equipped with an $\mathcal{X}$ structure that is compatible with composition (in place of a stable framing). For an $\mathcal{X}$ flow category $\mathcal{C}$, we obtain a coherent chain complex $K_{\mathcal{C}} \in \mathbf{Ch}(\mathrm{Mod}_{M\mathcal{X}})$, and, analogously to the realization in **Sp** (see realization), we define $|\mathcal{C}| \in \mathrm{Mod}_{\mathcal{X}}$ to be its iterated cofiber.

Our results

Let $\mathcal{X}$ be a stable tangential structure, let $\mathcal{C}$ be a $\mathcal{X}$ -structured flow category. An element $W_p = \{W_z\}_{|z|=p} \in E_1^{n,p}$ survives to the $(r+1)$ -th page if and only if it extends to a right flow module W of degree n over $\mathcal{C}_{[p,p-r]}$ (see Definition 3.28 for the formal definition).

To describe the differential, let W, V be a right and left flow module over $\mathcal{C}$ of degree n and $p-r$ respectively (see Definition 3.24, Definition 3.25). We define the punctured cube diagram

$$D : P(\{p, \dots, p-r\}) \setminus \{\emptyset\} \rightarrow \mathbf{Top},$$

where $P(-)$ denote the power set, by setting

$$D(\{a_k, \dots, a_1\}) = W_{a_k} \times \mathcal{M}_{a_k, a_{k-1}} \times \mathcal{M}_{a_{k-1}, a_{k-2}} \cdots \times \mathcal{M}_{a_2, a_1} \times V_{a_1},$$

where $a_k > a_{k-1} > \cdots > a_1$. The **Toda manifold** is the colimit which we denote

$$V \circ W := \text{colim } D$$

and we will see later that the Toda manifolds indeed inherits manifold structure from V, W , and that they are closed, $(n - p + r)$ -dimensional, $\mathcal{X}$ -structured.

For the spectral sequence, we consider a family of such constructions indexed by objects in $\mathcal{C}_{p-(r+1)}$. Specifically, for each $z \in \mathcal{C}_{p-(r+1)}$, we define a punctured cube diagram D_z by replacing the left module V with $\mathcal{M}_{-,z}$ in the general construction. This yields a Toda manifold

$$\mathcal{M}_{-,z} \circ W := \text{colim } D_z \in \Omega_{n-p+r}^{\mathcal{X}}.$$

Taking the disjoint union over all $z \in \mathcal{C}_{p-(r+1)}$, we obtain an element

$$\bigsqcup_{z \in \mathcal{C}_{p-(r+1)}} (\mathcal{M}_{-,z} \circ W) \in \bigoplus_{z \in \mathcal{C}_{p-(r+1)}} \Omega_{n-p+r}^{\mathcal{X}}.$$

Example. In Fig. 2, 3, and 4, we can see an example of an extension and its Toda manifold.

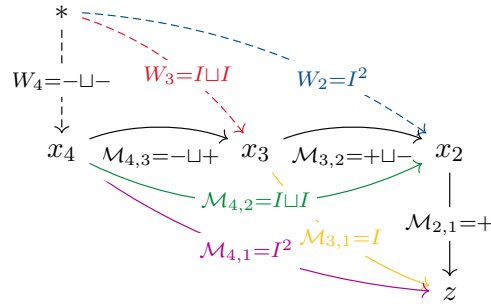


Figure 2: The solid part represents the flow category $\mathcal{C}$. Its objects are the x_i 's with $|x_i| = i$, and the arrows represent the morphism manifolds. The 0 dimensional framed manifold W_4 represents an element in $E_1^{4,4} = \Omega_{4-4}^{\text{fr}}$. The dashed arrows represent an right module over $\mathcal{C}_{[4,2]}$. The symbols $\pm, I$, etc., represent the flow manifolds themselves (where $\pm$ represents a framed point; the rest of framing is omitted for clarity).

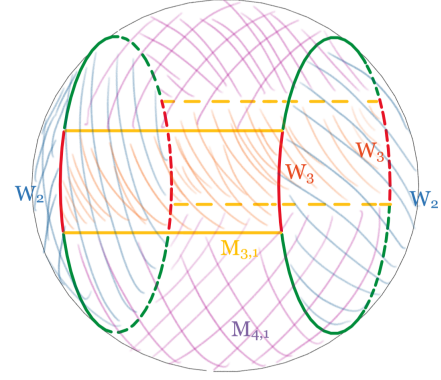
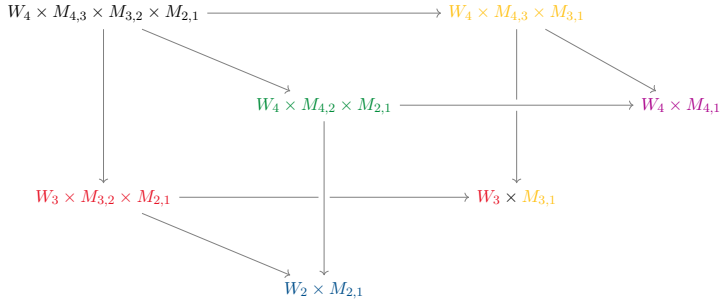


Figure 3: The punctured 3-cube diagram D_z associated with the object $z = x_1$ of degree 1, constructed from the right flow module W and the left flow module $\mathcal{M}_{-,z}$.

Figure 4: We can see that $\text{colim } D_z$ is a topological space that inherits a closed smooth manifold structure.

Theorem A. *Let $\mathcal{C}$ be a flow category with an $\mathcal{X}$ -tangential structure. There is a spectral sequence:*

$$E_1^{n,p} = \bigoplus_{|z|=p} \Omega_{n-p}^{\mathcal{X}}$$

which conditionally converges to the homotopy groups of the realization:

$$E_r^{n,p} \implies \pi_n |\mathcal{C}|.$$

An element

$$W_p = \{W_z\}_{|z|=p} \in E_1^{n,p}$$

survives to $E_{r+1}^{n,p}$ if and only if the collection W_p extends to a right flow module over $\mathcal{C}_{[p,p-r]}$ (see Definition 3.28). The differential is represented geometrically by the Toda manifolds:

$$d_{r+1}(W_p) = \bigcup_{y \in \mathcal{C}_{p-(r+1)}} (\mathcal{M}_{-,y} \circ W) \in \bigoplus_{y \in \mathcal{C}_{p-(r+1)}} \Omega_{n-p+r}^{\mathcal{X}} = E_{r+1}^{n-1,p-(r+1)}$$

****Note for rewriting.**** In Theorem A, the displayed Toda manifolds give classes in $\bigoplus_{y \in \mathcal{C}_{p-(r+1)}} \Omega_{n-p+r}^{\mathcal{X}} = E_1^{n-1,p-(r+1)}$. Their induced classes on E_{r+1} give d_{r+1} ; the full bordism group must not be identified with the later-page group. Explain independence from the chosen extension W at the page level.

A key tool used to prove Theorem A is a description of the spectral sequence associated with a coherent chain complex expressed in terms of chain complex maps.

Theorem B. *Let $K \in \text{Ch}(\mathbf{Sp})$ be a coherent chain complex in Spectra. Then there is a spectral sequence:*

$$E_1^{n,p} = \pi_{n-p} K_p,$$

which conditionally converges to

$$E_r^{n,p} \implies \pi_n \text{colim}_p (\mathcal{I}K)_p,$$

resolve: by Lurie we have that the S.S converge to the homotopy groups of the colim of the filtered object. We never justified why this colim is $|\mathcal{C}|$

move this ref after the thm?

Moreover, let $x : \mathbb{S}^{n-p} \rightarrow K_p$ represent an element of $E_1^{n,p}$. Then x survives to $E_{r+1}^{n,p}$ if and only if there exists a chain complex map:

$$\begin{array}{ccccccc} \mathbb{S}^{n-p} & \longrightarrow & 0 & \longrightarrow & \dots & \longrightarrow & 0 \\ \downarrow x & & \downarrow & & \downarrow & & \downarrow \\ K_p & \xrightarrow{d_p} & K_{p-1} & \xrightarrow{d_{p-1}} & \dots & \xrightarrow{d_{p-r}} & K_{p-r} \end{array} \quad (2)$$

Here, the top row is concentrated in degree p , and the bottom row is the restriction of K from degree p down to $p - r$. The differential $\partial_{r+1}(x)$ is given by the composition of chain maps:

$$\begin{array}{ccccc} \mathbb{S}^{n-p} & \longrightarrow & \dots & \longrightarrow & 0 \\ \downarrow x & & \downarrow & & \downarrow \\ K_p & \longrightarrow & \dots & \longrightarrow & K_{p-r} \\ \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & \dots & \longrightarrow & K_{p-r-1} \end{array}$$

The first map is precisely [Eq. \(2\)](#) and the second is induced by the structure of the chain complex K .

We obtain this result by combining the description of the spectral sequence of a filtered object in Lurie [19] with the results of [5].

Remark 1.1. If the flow category $\mathcal{C}$ (or the chain complex K) is bounded from below, the conditional convergence in [Theorem A](#) and [Theorem B](#) upgrades to strong convergence (referred to simply as convergence in [19, Prop. 1.2.2.14]). In general, the stable filtered object associated with a chain complex is complete, which allows us to use the Milnor exact sequence to satisfy Boardman’s completeness criteria [7, Def. 5.10] and guarantee conditional convergence even when unbounded (see [Section 2.2](#) for details).

Relation to other work

This paper contributes to the broader project of providing a geometric model for spectra (see [1]), building on the seminal work of Cohen, Jones, and Segal [9] who first realized Morse and Floer data as stable homotopy types.

In low-dimensional topology, flow categories are used to lift Khovanov homology to a stable homotopy type [15], where Pontryagin-Thom gluing of 1-dimensional moduli spaces computes the secondary cohomology operation Sq^2 [16]. Our work generalizes this gluing to higher-dimensional flow manifolds with corners to geometrically model all higher spectral sequence differentials and Toda brackets.

In symplectic geometry, flow categories and modules arise in Lagrangian Floer theory to construct a spectral Fukaya category over the sphere spectrum [26], where successive truncations of flow modules act as obstructions to lifting quasi-isomorphisms. A natural direction for future work is generalizing our spectral sequence framework to equivariant flow categories, such as the Morse-Bott models of Côté–Kartal [11] and the equivariant flow categories of Rezchikov [27] used to construct cyclotomic structures on Floer spectra.

Morphisms and invariance. In [1], the authors construct an ∞ -category Flow^{fr} (see also [26] for the homotopy category) and show that it is equivalent to $\mathbf{Sp}$. The morphisms in this category are *flow bimodules*. Given two Morse-Smale functions f and g on a compact manifold M , one obtains two framed flow categories, $\mathcal{C}_f$ and $\mathcal{C}_g$, and a flow bimodule between them, showing that the stable homotopy type is independent of the choice of Morse-Smale function. However, the different CW structures yield different spectral sequences, so the spectral sequence itself is not an invariant of the equivalence class in Flow^{fr} . To address this, we expect there is a construction analogous to that in [1] that defines another ∞ -category with a modified class of morphisms. Specifically, for a 2-morphism R between flow categories $\mathcal{C}$ and $\mathcal{D}$, we require that if $\|y\| > \|x\|$, then $R_{x,y} = \emptyset$. This construction would provide a flow-categorical model for the subcategory of $\{K_p\} \in \text{Ch}(\mathbf{Sp})$ satisfying $K_p = \bigoplus \mathbb{S}$. In this setup, the two Morse-Smale functions on a manifold M produce distinct objects, and we can think of Theorem A as a spectral sequence of a filtered flow category $F \in \widehat{\text{Fil}}(\text{Flow}^{\text{X}})$ induced by the index filtration.

Remark 1.2. In this paper, we do not attempt to implement this full theoretical framework. The complete ∞ -categorical structure is unnecessary for our purposes; instead, we only require a practical model for the composition of left and right flow modules to extract the geometric differentials of the spectral sequence.

The interpretation of gluing manifolds as Toda brackets or Massey products in cobordism appears in [2] (see also [23]), where the Massey product is defined via the gluing of manifolds with boundary.

Theorem B bears a similar flavor to other works on spectral sequences associated with various objects. In [5], the authors explain the connection between filtered objects and coherent chain complexes, and they also define a spectral sequence for coherent chain complexes using the Beilinson t -structure. We also recommend [4] as an introduction to spectral sequences and their relation both to chain complexes and to the spectral sequence of filtered objects defined in [19]. The construction of the spectral sequence in this paper is similar to that in [6], where the authors define a spectral sequence associated with a simplicial object.

Paper layout

Section 2 is devoted to the framework of chain complexes in (stable) ∞ -categories and contains the proof of Theorem B.

Section 3 recalls the definitions of manifolds with corners, framed flow categories, and the Pontryagin-Thom construction for manifolds with corners. At the end of this section, we demonstrate several properties of the Pontryagin-Thom construction that will be useful in later sections.

Section 4 establishes a correspondence between chain complexes in the category of spectra (with values $\bigoplus \mathbb{S}$) and framed flow categories.

Our goal in Sections 5 and 6 is to upgrade this correspondence so that it interacts naturally with spectral sequences. In other words, given a framed flow category, we show how to produce a spectral sequence for the corresponding chain complex and provide a geometrical description of its pages and differentials.

In Section 5, we compare various models for chain complexes by considering different models for ∞ -categories (namely, topologically enriched, simplicially enriched, and simplicial sets). This comparison allows us to give a straightforward description of the composition of maps between flow categories.

Section 6 translates this description of composition into a geometrical framework and establishes the existence of a spectral sequence associated with framed flow categories.

The final section explains how the framing structure can be replaced by a more general tangential structure, and discusses how different structures interact. This section also contains examples of spectral sequences arising from flow categories.

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2 Chains Complexes and their Spectral Sequences

Spectral sequences for filtered objects in Spectra are treated in Lurie’s work [19]. A key insight, developed by Ariotta and Krause [5], is that within a stable ∞ -category, filtered objects are equivalent to coherent chain complexes. Building upon this equivalence, this section aims to provide an explicit description of the spectral sequence associated with a coherent chain complex.

2.1 Chain Complexes and Filtered Objects

In this section, let $\mathcal{C}$ be a stable ∞ -category with sequential limits and colimits. Following [5], we will introduce the categories of chain complexes and filtered objects in ∞ -category. We will show how to connect between them and the equivalence between the two categories.

Remark 2.1. For being compatible with flow categories, our chain complexes will decrease and hence the filtered objects will increase, as oppose to [5].

Definition 2.2. We define the category Ch to be the pointed (ordinary) category with

$$\begin{aligned} \text{ob } \text{Ch} &:= \mathbb{Z} \cup \{*\} \\ \text{Hom}_{\text{Ch}}(i, j) &:= \begin{cases} \{\text{id}, \mathbf{0}\} & i = j \\ \{d_i, \mathbf{0}\} & i = j + 1 \\ \{\mathbf{0}\} & \text{otherwise} \end{cases} \end{aligned}$$

where $*$ is the zero object and $\mathbf{0}$ is the map $i \rightarrow * \rightarrow j$.

For $\mathcal{C}$ as above, we define *chain complexes in $\mathcal{C}$* (or just *chains*) to be the full subcategory

$$\text{Ch}(\mathcal{C}) := \text{Fun}^*(\text{Ch}, \mathcal{C}) \subset \text{Fun}(\text{Ch}, \mathcal{C})$$

of pointed functors.

We denote by Ch_b the full subcategory of $\text{Ch}(\mathcal{C})$ spanned by the objects K such that there exists $N \in \mathbb{N}$ with:

$$K_i = 0 \text{ for all } i < N.$$

We call such chain complexes *bounded below*.

Remark. If we don’t mention otherwise, we assume that $K \in \text{Ch}_b$ is bounded below by $N = 0$.

Example. The category $\text{Ch}(\mathbf{Ab})$ of chain complexes in abelian groups is the usual category of chain complexes.

Remark 2.3. In the case where $\mathcal{C}$ is an ∞ -category, the reader may want to think of $\text{Ch}(\mathcal{C})$ as chain complexes, with null-homotopies of the composition of two maps along with higher null-homotopies: given a composition of three maps in the chain complex

$$K_p \rightarrow K_{p-1} \rightarrow K_{p-2} \rightarrow K_{p-3},$$

we get two null-homotopies of the map $K_p \rightarrow K_{p-3}$, which give us a map $S^1 \rightarrow \text{Hom}(K_p, K_{p-3})$. Part of the data of a chain complex is a null-homotopy of this loop and similarly higher null-homotopies for higher loops.

Definition 2.4. We denote by $\mathbb{Z}$ the category of integers with the usual increasing order.

Definition 2.5. Let $\mathcal{C}$ be a stable ∞ -category with sequential limits and colimits. We define the category of (increasing) complete *filtered objects in $\mathcal{C}$* to be the full subcategory

$$\widehat{\text{Fil}}(\mathcal{C}) \subset \text{Fun}(\mathbb{Z}, \mathcal{C})$$

spanned by functors F such that $\lim_p F_p = 0$. We define the **bounded below filtered objects** to be the full subcategory

$$\widehat{\text{Fil}}_{\text{b}}(\mathcal{C}) \subset \widehat{\text{Fil}}(\mathcal{C})$$

spanned by functors F such that there exists $N \in \mathbb{N}$ with $F_p = 0$ for $p < N$.

Note that the image of the stable Yoneda embedding

$$\mathfrak{Y}_{(-)} : \mathbb{Z}^{\text{op}} \rightarrow \text{Fun}(\mathbb{Z}, \mathbf{Sp}).$$

is given by

$$\mathfrak{Y}_p(q) = \text{Map}_{\mathbf{Sp}}(p, q) = \begin{cases} \mathbb{S} & q \geq p \\ 0 & q < p \end{cases}$$

Let $p \rightarrow (p-1)$ be a morphism in $\mathbb{Z}^{\text{op}}$, the morphisms $\mathfrak{Y}_p \rightarrow \mathfrak{Y}_{p-1}$ is given by

$$\begin{array}{ccccccc} \dots & \longrightarrow & 0 & \longrightarrow & 0 & \longrightarrow & \mathbb{S} \longrightarrow \dots \\ & & \downarrow & & \downarrow & & \downarrow \\ \dots & \longrightarrow & 0 & \longrightarrow & \mathbb{S} & \longrightarrow & \mathbb{S} \longrightarrow \dots \end{array}$$

Definition 2.6. For $n, p \in \mathbb{Z}$ we define the following chain complex $\mathbb{S}_p^n \in \text{Ch}(\mathbf{Sp})$ by

$$\mathbb{S}_p^n(q) := \begin{cases} \mathbb{S}^n & q = p \\ 0 & \text{otherwise} \end{cases}$$

Note that there is the map in $\text{Ch}(\mathbf{Sp})$

$$Y_{n,p} : \mathbb{S}_p^{n-1} \rightarrow \mathbb{S}_{p-1}^n$$

which visually given by

$$\begin{array}{ccccccc} \dots & \longrightarrow & 0 & \longrightarrow & \mathbb{S}^{n-1} & \longrightarrow & 0 \longrightarrow 0 \longrightarrow \dots \\ & & \downarrow & & \downarrow & & \downarrow \\ \dots & \longrightarrow & 0 & \longrightarrow & 0 & \longrightarrow & \mathbb{S}^n \longrightarrow 0 \longrightarrow \dots \end{array}$$

which is defined using the fact that $\mathbb{S}^n \simeq \Sigma \mathbb{S}^{n-1}$ in $\mathbf{Sp}$.

In particular we have a map

$$Y_p := Y_{-(p-1),p} : \mathbb{S}_p^{-p} \rightarrow \mathbb{S}_{p-1}^{-(p-1)}$$

Definition 2.7. We define

$$\mathcal{A}: \widehat{\text{Fil}}(\mathbf{Sp}) \rightarrow \text{Ch}(\mathbf{Sp})$$

to be the colimit preserving functor that determined by mapping

$$\mathcal{A}(\mathcal{J}_p) := \mathbb{S}_p^{-p}$$

and the morphisms are given by Y_p .

We denote it's right adjoint

$$\widehat{\text{Fil}}(\mathbf{Sp}) \leftarrow \text{Ch}(\mathbf{Sp}) : \mathcal{I}$$

In [5], Ariotta define these functors and proves

Theorem 2.8 ([5]). *Let $\mathcal{C}$ be a stable ∞ -category with sequential limits and colimits. We have an equivalence*

$$\mathcal{A} : \widehat{\text{Fil}}(\mathcal{C}) \simeq \text{Ch}(\mathcal{C}) : \mathcal{I}.$$

An immediate consequence is the representability of the evaluation:

$$\text{Map}(\mathbb{S}_p^{-p}, \mathcal{A}F) \simeq \text{Map}(\mathcal{I}\mathbb{S}_p^{-p}, F) \simeq \text{Map}(\mathcal{J}_p, F) \simeq F_p, \quad (3)$$

which give us a formula for the p -th piece of $\mathcal{I}K$:

$$(\mathcal{I}K)_p \simeq \text{Map}(\mathbb{S}_p^{-p}, K).$$

Definition 2.9. Let $\mathcal{C}$ be a stable ∞ -category and $p \in \mathbb{Z}$, we write $\iota_{[p-1, p]} : \{(p-1) \rightarrow p\} \rightarrow \mathbb{Z}$ for the inclusion functor and

$$(\iota_{[p-1, p]})^* : \text{Fun}(\mathbb{Z}, \mathcal{C}) \rightarrow \text{Fun}(\Delta^1, \mathcal{C})$$

for the restriction (pre-composition) along $\iota_{[p-1, p]}$. We define the p -th associated graded functor by:

$$\text{gr}_p := \text{cof} \circ (\iota_{[p-1, p]})^* : \text{Fun}(\mathbb{Z}, \mathcal{C}) \rightarrow \mathcal{C},$$

so that

$$F \mapsto F_p / F_{p-1}.$$

The map

$$F_p / F_{p-1} \rightarrow \Sigma F_{p-1} / F_{p-2}$$

given by the connecting morphism of the cofiber, and then taking quotient, induces a transformation:

$$\text{gr}_p \rightarrow \Sigma \text{gr}_{p-1}.$$

A direct calculation shows that

$$\Sigma^{-p} \text{gr}_q(\mathcal{J}_p) = (\mathcal{A}\mathcal{J}_p)_q$$

and we deduce that

Corollary 2.10.

$$(\mathcal{A}-)_p \simeq \Sigma^{-p} \text{gr}_p(-)$$

FILLLL

As functors

$$\widehat{\text{Fil}}(\mathbf{Sp}) \rightarrow \mathbf{Sp}$$

We now provide an explicit expression for $\mathcal{I}$.

First, we define a the following chain complex in $\mathbf{Sp}$,

$$\Sigma^n y_p := \begin{cases} \mathbb{S}^n & q \in \{p, p-1\} \\ 0 & \text{otherwise} \end{cases}$$

In particular we have

$$\mathrm{cof}(Y_p) \simeq \Sigma^{-(p-1)} y_p$$

On the other hand (see [5] Lemma 3.12), one can show that the evaluation functor

$$(-)_p: \mathrm{Ch}(\mathbf{Sp}) \rightarrow \mathbf{Sp}, \quad K \mapsto K_p$$

is represented:

$$\mathrm{Map}(y_p, -) \simeq (-)_p.$$

The objects in the following fiber sequence

$$\mathbb{S}_p^{-p} \rightarrow \mathbb{S}_{p-1}^{-(p-1)} \rightarrow \mathrm{cof}(Y_p) \quad (4)$$

represents the functors

$$(\mathcal{I}-)_p, \quad (\mathcal{I}-)_{p-1}, \quad \Sigma^{-(p-1)}(-)_p.$$

By taking the contravariant Map functor to Eq. (4) we get the following result:

Lemma 2.11. *There is a cofiber sequence of functors $\mathrm{Ch}(\mathbf{Sp}) \rightarrow \mathbf{Sp}$,*

$$\Sigma^{-(p-1)}(-)_p \rightarrow (\mathcal{I}-)_{p-1} \rightarrow (\mathcal{I}-)_p \quad (5)$$

****Note for rewriting.**** Check the suspension in Lemma 2.11: since $K_p \simeq \Sigma^{-p}(F_p/F_{p-1})$, rotating $F_{p-1} \rightarrow F_p \rightarrow F_p/F_{p-1}$ gives $\Sigma^{p-1}K_p \rightarrow F_{p-1} \rightarrow F_p$. The displayed $\Sigma^{-(p-1)}K_p$ has the wrong shift; reconcile this with the representing sequence above.

A simple corollary is

Corollary 2.12. *Let $K \in \mathrm{Ch}_b(\mathbf{Sp})$ bounded below by N . Then for $p \leq N$ we have*

$$(\mathcal{I}K)_p = 0$$

****Note for rewriting.**** Correct the endpoint of this bounded-below claim: if $K_i = 0$ for $i < N$, then $(\mathcal{I}K)_p = 0$ for $p < N$, while $(\mathcal{I}K)_N \simeq \Sigma^N K_N$ in general. Thus the displayed assertion with $p \leq N$ is too strong. I think that we need to change $p \leq N$ to $p < N$ in the statement of the corollary.

which obtained by inductively applying Lemma 2.11.

Example. Let

$$\cdots \rightarrow 0 \rightarrow F_0 \rightarrow F_1 \rightarrow F_2 \rightarrow F_3 \rightarrow 0 \rightarrow \cdots$$

be a filtered object in $\mathbf{Sp}$. Then $K := \mathcal{A}F$ is given by:

$$\cdots \rightarrow 0 \rightarrow \Sigma^{-3}F_3/F_2 \rightarrow \Sigma^{-2}F_2/F_1 \rightarrow \Sigma^{-1}F_1/F_0 \rightarrow F_0 \rightarrow 0 \cdots$$

where the zeros on the left are F_3/F_3 . The composition

$$K_3 = \Sigma^{-3}F_3/F_2 \rightarrow \Sigma^{-2}F_2/F_1 \rightarrow \Sigma^{-1}F_1/F_0 = K_1$$

can be factored as:

$$\Sigma^{-3}F_3/F_2 \rightarrow \Sigma^{-2}F_2 \rightarrow \Sigma^{-2}F_2/F_1 \rightarrow \Sigma^{-1}F_1 \rightarrow \Sigma^{-1}F_1/F_0,$$

where the middle part is a cofiber sequence; hence, the composition is null-homotopic. Following [Remark 2.3](#), this null-homotopy is the "non-higher" part of the chain complex K .

To recover $F = \mathcal{I}K$ from K , we can use [Lemma 2.11](#). The fact that K is bounded below by -1 yields that

$$(\mathcal{I}K)_p = 0 \quad \text{for } p \leq -1$$

From that following cofiber sequence We can see that $F_0 = K_0$

$$\Sigma^{-(-1)}K_0 \rightarrow \overbrace{F_{-1}}^{=0} \rightarrow F_0.$$

We can recover F_1 as the cofiber

$$K_1 \rightarrow \overbrace{F_0}^{=K_0} \rightarrow F_1.$$

And by taking F_2 is reconstructed as an iterated cofiber, from the cofiber sequence :

$$\Sigma^{-1}K_2 \rightarrow \overbrace{F_1}^{=K_1/K_0} \rightarrow F_2.$$

Remark 2.13. An "non example" is given by taking a topological space X and the filtered space $F_p := \text{sk}_p X$. If X is finite, then there is an $N \in \mathbb{N}$ - an upper bound for the filtration. In this case, we can define the following chain complex in **Spaces**

$$K_p := \Sigma^{N-p} \text{gr}_p F = \text{sk}_p X / \text{sk}_{p-1} X \simeq \bigvee_{p \text{ cells}} S^N$$

and the morphisms $\text{gr}_p \rightarrow \Sigma \text{gr}_{p-1}$ that induces the maps $K_p \rightarrow K_{p-1}$, are given by the maps:

$$\bigvee S^p = \text{sk}_p X / \text{sk}_{p-1} X \xrightarrow{\partial_p} \text{sk}_{p-1} X / \text{sk}_{p-2} X = \Sigma \bigvee S^{p-1},$$

which are more commonly known as the differentials in the chain complex that computes the cellular homology of X . By taking iterated cofibers we can recover F only up to suspension, for example

$$\Sigma^N F_1 \simeq \text{cof}(\Sigma^{N-1} F_1 / F_0 \rightarrow S^N F_0)$$

2.2 Spectral Sequences of Chain Complexes

We will start with the construction of spectral sequence of a filtered object, following the notations and definitions in [\[20\]](#). Combined with the last subsection we will get a description of the pages of a spectral sequence of a chain complex. The main output is [Proposition 2.24](#).

For our discussion, we assume that $\mathcal{C}$ is a stable ∞ category equipped with a t -structure $(\mathcal{C}^{\leq 0}, \mathcal{C}^{\geq 0})$. We begin by defining the spectral sequence associated with a filtered object in $\mathcal{C}$.

Definition 2.14. [19, Def. 1.2.2.9] Let F be a filtered object in $\mathcal{C}$, i.e., a sequence of morphisms $\{F_p\}_{p \in \mathbb{Z}}$ such that

$$\cdots \rightarrow F_{p-1} \rightarrow F_p \rightarrow F_{p+1} \rightarrow \cdots. \quad (6)$$

For each $p \in \mathbb{Z}$, define the graded pieces by

$$\mathrm{gr}_p F = \mathrm{cof}(F_{p-1} \rightarrow F_p). \quad (7)$$

We define a spectral sequence $\{E_r^{n,p}, d_r^{n,p}\}$ associated with F by setting

$$E_r^{n,p} = \mathrm{Im}(\pi_n(F_p/F_{p-r}) \rightarrow \pi_n(F_{p+r-1}/F_{p-1})), \quad (8)$$

where the map is induced by the compositions of the inclusion and projection:

$$F_p/F_{p-r} \rightarrow F_p/F_{p-1} \rightarrow F_{p+r-1}/F_{p-1}. \quad (9)$$

The differentials $d_r^{n,p}: E_r^{n,p} \rightarrow E_r^{n-1,p-r}$ are determined by the connecting morphisms via the commutativity of the diagram:

$$\begin{array}{ccccc} \pi_n(F_p/F_{p-r}) & \longrightarrow & E_r^{n,p} & \longrightarrow & \pi_n(F_{p+r-1}/F_{p-1}) \\ \downarrow \delta_{p-r} & & \downarrow d_r^{n,p} & & \downarrow \delta_{p-1} \\ \pi_{n-1}(F_{p-r}/F_{p-2r}) & \longrightarrow & E_r^{n-1,p-r} & \longrightarrow & \pi_{n-1}(F_{p-1}/F_{p-r-1}), \end{array} \quad (10)$$

where the vertical maps δ_{p-r} and δ_{p-1} are the connecting homomorphisms from the long exact sequences of homotopy groups associated with the cofiber sequences:

$$F_{p-r} \rightarrow F_p \rightarrow F_p/F_{p-r} \quad \text{and} \quad F_{p-1} \rightarrow F_{p+r-1} \rightarrow F_{p+r-1}/F_{p-1}. \quad (11)$$

Theorem 2.15. [19, Prop. 1.2.2.14] Let $\mathcal{C}$ be a stable ∞ -category equipped with a t -structure, that admits sequential colimits and that the t -structure is compatible with sequential colimits. Let F be a bounded below filtered object in $\mathcal{C}$ (see Definition 2.5). Then the spectral sequence $\{E_r^{n,p}, d_r^{n,p}\}$ (see Definition 2.14) converges to the homotopy groups of the colimit of F :

$$E_r^{n,p} \Rightarrow \pi_n \operatorname{colim}_p F_p.$$

Example 2.16. Consider a CW complex X with its skeletal filtration $\{\mathrm{sk}_p X\}_{p \geq 0}$. Applying the suspension spectrum functor Σ^∞ , we obtain a filtered spectrum F in $\mathbf{Sp}$, where $F_p = \Sigma^\infty \mathrm{sk}_p X$. Let $K = \mathcal{A}F$ be the associated chain complex (as in Remark 2.13). The spectral sequence arising from this filtration has first page:

$$E_1^{n,p} = \pi_n \mathrm{gr}_p F = \pi_n \left(\bigoplus \mathbb{S}^p \right) = \pi_{n-p} K_p,$$

since each $\mathrm{gr}_p F$ is a wedge of spheres suspended p times, and K_p is the p -th term in the chain complex K .

This spectral sequence converges to:

$$\pi_n \Sigma^\infty X,$$

which are the stable homotopy groups of the space X .

More generally, for any spectrum A , we can consider the filtered spectrum $F_p \otimes A$, obtaining a spectral sequence converging to the generalized homology:

$$A_n(X).$$

This spectral sequence is known as the *Atiyah–Hirzebruch spectral sequence*.

Definition 2.17. Let F be a filtered object in stable ∞ -category $\mathcal{C}$ and let $E_r^{n,p}$ be it's spectral sequence. An element $x \in E_1^{n,p}$ is said to **survive to the r -th page** if it lies in the image of the map:

$$x \in \text{Im}(\pi_n(F_p/F_{p-r}) \rightarrow \pi_n(F_p/F_{p-1})).$$

We denote the collection of such elements by $\mathcal{E}_r^{n,p}$. If $x' \in \pi_n(F_p/F_{p-r})$ maps to x , we say that x' **lifts** x .

Intuitively, an element survives to the r -th page if it is not killed by the differentials d_k for $1 \leq k < r$.

Remark 2.18. In the stable homotopy category $\mathbf{Sp}$, the homotopy groups π_n are representable by spheres $\mathbb{S}^n$. Thus, elements in

$$E_1^{n,p} = \pi_n \text{gr}_p F = \pi_{n-p} K_p$$

are represented by maps $x: \mathbb{S}^{n-p} \rightarrow K_p$.

We introduce a notation for subcategories of chain complexes.

Definition 2.19. For integers $a \leq b$, let $\text{Ch}_{[b,a]}$ denote the full subcategory of the poset category Ch spanned by the objects

$$\{i \in \mathbb{Z} \mid b \geq i \geq a\} \cup \{*\}, \quad (12)$$

where $*$ denotes the initial object. Given a pointed category $\mathcal{C}$, we define

$$\text{Ch}_{[b,a]}(\mathcal{C}) = \text{Fun}^*(\text{Ch}_{[b,a]}, \mathcal{C}) \subseteq \text{Fun}(\text{Ch}_{[b,a]}, \mathcal{C}), \quad (13)$$

the full subcategory of pointed functors from $\text{Ch}_{[b,a]}$ to $\mathcal{C}$.

Precomposition with the inclusion $\text{Ch}_{[b,a]} \hookrightarrow \text{Ch}$ induces restriction functors:

$$i_{[b,a]}^*: \text{Ch}(\mathcal{C}) \rightarrow \text{Ch}_{[b,a]}(\mathcal{C}).$$

Restrictions from below

$$(i_{\geq a})^*: \text{Fun}(\text{Ch}, \mathcal{C}) \rightarrow \text{Fun}(\text{Ch}_{\geq a}, \mathcal{C}) =: \text{Ch}_{\geq a}(\mathcal{C}) \quad (14)$$

admits the following right adjoint:

$$(i_{\geq a})_*: \text{Ch}_{\geq a}(\mathbf{Sp}) \rightarrow \text{Ch}(\mathbf{Sp})$$

which given by right padding with zeros:

$$(i_{\geq a})_* K(p) = \begin{cases} K(p) & p \geq a \\ 0 & \text{otherwise} \end{cases} \quad (15)$$

On the other hand, restriction from above have padding as left adjoint:

$$(i_{\leq b})_! K(p) = \begin{cases} 0 & p > b \\ K(p) & b \geq p \end{cases} \quad (16)$$

We denote the composition by:

$$K \mapsto K_{[b,a]} =: (i_{\leq b})_! (i_{\geq a})_* K$$

We can do the same for filtered object.

Definition 2.20. For two integers $a \leq b$ we define

$$\text{Fil}_{[a,b]}(\mathbf{Sp}) := \text{Fun}([a,b], \mathbf{Sp})$$

Note that the inclusion

$$i_{\leq n} : (-\infty, n] \rightarrow \mathbb{Z}$$

induces the restriction functor:

$$i_{\leq n}^* : \text{Fil}(\mathbf{Sp}) \rightarrow \text{Fil}_{(-\infty, n]}(\mathbf{Sp}) =: \text{Fil}_{\leq n}(\mathbf{Sp})$$

and left adjoint

$$(i_{\leq n})_! : \text{Fil}_{\leq n}(\mathbf{Sp}) \rightarrow \text{Fil}(\mathbf{Sp})$$

which is given by

$$(i_{\leq n})_! F(p) = \begin{cases} F(p) & p \leq n \\ F(n) & \text{otherwise} \end{cases}$$

Similarly we define

$$\text{Fil}_{\geq n}(\mathbf{Sp})$$

The map

$$i_{\geq n} : [n, \infty) \rightarrow \mathbb{Z}$$

induces the restriction

$$i_{\geq n}^* : \text{Fil}(\mathbf{Sp}) \rightarrow \text{Fil}_{\geq n}(\mathbf{Sp})$$

and it's right adjoint

$$(i_{\geq n})_* F(p) = \begin{cases} F(p) & n \leq p \\ 0 & \text{otherwise} \end{cases}$$

Note. sometimes we will identify between $F \in \widehat{\text{Fil}}_{[n,m]}(\mathbf{Sp})$ to it's image in $\widehat{\text{Fil}}(\mathbf{Sp})$.

On the other hand we can define the "quotient truncation" functor.

Definition 2.21. Let $n \in \mathbb{Z}$ we define

$$q_n : \text{Fil}(\mathbf{Sp}) \rightarrow \text{Fil}_{\geq n}(\mathbf{Sp})$$

by

$$(q_n F)_p := \begin{cases} F_p / F_n & n+1 \leq p \\ 0 & \text{otherwise} \end{cases}$$

This functor behave nicely with the equivalence to chain complexes:

Corollary 2.22. *Let $F \in \widehat{\text{Fil}}(\mathbf{Sp})$, we have that:*

$$\mathcal{A}q_n F \simeq i_{\geq n+1}^* \mathcal{A}F$$

treated as objects in $\text{Ch}_{[\infty, n+1]}(\mathbf{Sp})$ or $\text{Ch}(\mathbf{Sp})$.

Proof. Both constructions preserve colimits, so we compare them on the filtered Yoneda generators $\mathcal{A}\mathbb{S}_j$ and their transition maps. If $j \leq n$, then $q_n \mathcal{A}\mathbb{S}_j = 0$, and restriction also kills $\mathcal{A}\mathbb{S}_j = \mathbb{S}_j^{-j}$. If $j > n$, quotient truncation leaves $\mathcal{A}\mathbb{S}_j$ unchanged, while restriction leaves $\mathbb{S}_j^{-j}$ unchanged. These identifications preserve the transition maps between retained generators; every transition with a discarded target becomes the unique map to zero. Thus they agree on the generator diagram, including its compositions, and extend by colimits to a natural equivalence of coherent chain complexes. $\square$

Lemma 2.23. *Let $K \in \text{Ch}(\mathbf{Sp})$ and $F = \mathcal{I}K$ its equivalent filtered spectrum, then*

$$\text{Map}(\mathbb{S}_p^{-p}, i_{\geq n+1}^* K) \simeq F_p / F_n$$

Proof. Note that $\mathcal{A}F = K$. We use the corollary above and [Eq. \(3\)](#) to get:

$$\text{Map}(\mathbb{S}_p^{-p}, i_{\geq n+1}^* K) \simeq \text{Map}(\mathbb{S}_p^{-p}, \mathcal{A}q_n F) \simeq (q_n F)_p = F_p / F_n.$$

$\square$

Let K be a chain complex, and let $F = \mathcal{I}K$, then by passing Lurie's construction of spectral sequence of F we get the following

Proposition 2.24. *Let $K \in \text{Ch}(\mathbf{Sp})$ be a chain complex in Spectra, let $F = \mathcal{I}K$ the equivalent filtered spectrum and $E_r^{n,p}$ its spectral sequence (see [Definition 2.14](#) and [Theorem 2.15](#)). By passing Lurie's construction we get that*

$$E_1^{n,p} = \pi_{n-p} K_p,$$

Moreover, let $x: \mathbb{S}^{n-p} \rightarrow K_p$ represent an element in $E_1^{n,p}$. Then x survives to E_r if and only if there are maps of chain complexes:

$$\mathbb{S}_p^{n-p} \rightarrow K_{[p, p-r+1]}$$

such that the restriction

$$\mathbb{S}_p^{n-p} \rightarrow K_{[p, p-r+1]} \rightarrow i_{\geq p}^* K_{[p, p-r+1]} = K_p$$

is x .

We depict such morphism as

$$\begin{array}{ccccccc} \mathbb{S}^{n-p} & \longrightarrow & 0 & \longrightarrow & \dots & \longrightarrow & 0 \\ \downarrow x & & \downarrow & & \downarrow & & \downarrow \\ K_p & \xrightarrow{d_p} & K_{p-1} & \xrightarrow{d_{p-1}} & \dots & \xrightarrow{d_{p-r+2}} & K_{p-r+1}. \end{array}$$

Proof. The first page is given in [Remark 2.18](#). Similar to $i_{\geq n}$, the inclusion

$$j := i_{\geq p-1, p-r} : (\infty, p-1] \rightarrow (\infty, p-r]$$

induces the restriction functor j^* . By post composing we get a map

$$\mathrm{Map}(\mathbb{S}_p^{-p}, i_{\geq p-r+1}^* K) \xrightarrow{j^* \circ} \mathrm{Map}(\mathbb{S}_p^{-p}, i_{\geq p}^* K).$$

By suspending both sides (from $\mathbb{S}_p^{-p}$ to $\mathbb{S}_p^{n-p}$) we get the map

$$\mathrm{Map}(\mathbb{S}_p^{n-p}, i_{\geq p-r+1}^* K) \xrightarrow{j^* \circ} \mathrm{Map}(\mathbb{S}_p^{n-p}, i_{\geq p}^* K). \quad (17)$$

To get that visualization in the proposition, we first clarify that we use $\mathbb{S}_p^{n-p}$ for both the restricted and the usual versions of the chain that concentrated at p and we can write:

$$\begin{aligned} & \mathrm{Map}_{[p, p-r+1]}(\mathbb{S}_p^{n-p}, K_{[p, p-r+1]}) \\ &= \mathrm{Map}_{[p, p-r+1]}(\mathbb{S}_p^{n-p}, i_{\leq p}^* i_{\geq p-r+1}^* K) \simeq \mathrm{Map}_{(\infty, p-r+1]}((i_{\leq p})! \mathbb{S}_p^{n-p}, i_{\geq p-r+1}^* K) \\ &= \mathrm{Map}_{(\infty, p-r+1]}(\mathbb{S}_p^{n-p}, i_{\geq p-r+1}^* K) \simeq \Sigma^{n-p} F_p / F_{p-r} \end{aligned}$$

****Note for rewriting.**** Check the final suspension in the preceding display: the mapping spectrum out of $\mathbb{S}_p^{n-p}$ should be $\Sigma^{-n}(F_p/F_{p-r})$, as in [Eq. \(18\)](#), rather than the displayed $\Sigma^{n-p}(F_p/F_{p-r})$.

where the equalities are notations and the last equivalence is [Lemma 2.23](#). We can do the same to the right hand side of [Eq. \(17\)](#) and get the map:

$$\Sigma^{-n} F_p / F_{p-r} \simeq \mathrm{Map}(\mathbb{S}_p^{n-p}, K_{[p, p-r+1]}) \xrightarrow{j^* \circ} \mathrm{Map}(\mathbb{S}_p^{n-p}, K_p) \simeq \Sigma^{-n} F_p / F_{p-1} \quad (18)$$

By taking π_0 we conclude that $x \in \pi_n F_p / F_{p-1}$ is in the image of $\pi_n F_p / F_{p-r}$ if and only if the representing map

$$x : \mathbb{S}^n \rightarrow K_p$$

factor through $K_{[p, p-r+1]}$, which conclude the proof. $\square$

Definition 2.25. A spectral sequence $\{E_r^{n,p}, d_r^{n,p}\}$ associated with a filtered object F converges conditionally to the homotopy groups of $\mathrm{colim}_p F_p$ if F is exhaustive and complete. Specifically, following Boardman [\[7, Def. 5.10\]](#), completeness requires that both the limit and the derived limit over the filtering vanish. In the ∞ -categorical setting, this is equivalent to the single condition $\lim_p F_p \simeq 0$.

Remark 2.26. For any coherent chain complex K , the associated filtered object $F = \mathcal{I}K$ is complete by construction.

Corollary 2.27. Let $K \in \mathrm{Ch}(\mathbf{Sp})$ be a coherent chain complex and $F = \mathcal{I}K$ its equivalent filtered spectrum.

1. The associated spectral sequence converges conditionally to $\pi_* \mathrm{colim}_p F_p$.
2. If K is bounded below, then the spectral sequence converges strongly to $\pi_* \mathrm{colim}_p F_p$.

Proof. For the first statement, since F is complete ($\lim_p F_p \simeq 0$), the Milnor exact sequence applied to the sequential limit of F_p implies that the $\lim^1$ term also vanishes. This satisfies the precise criteria for conditional convergence [7, Def. 5.10]. For the second statement, if K is bounded below, then F is a bounded below filtered object. By Theorem 2.15, this bounded below condition promotes conditional convergence to strong convergence. $\square$

****Note for rewriting.**** Check the unbounded conditional-convergence assertion and the use of Boardman’s completeness criteria against Lurie’s spectral-sequence discussion [19]. In particular, verify the exact hypotheses and conclusion before giving this claim the same prominence as the bounded-below case.

Example 2.28. Returning to Example 2.16, consider an element $x: \mathbb{S}^{n-p} \rightarrow K_p$ representing a class in $E_1^{n,p}$. The differential $d_1^{n,p}: E_1^{n,p} \rightarrow E_1^{n-1,p-1}$ is given by:

$$d_1^{n,p}(x) = d_p \circ x, \quad (19)$$

where $d_p: K_p \rightarrow K_{p-1}$ is the differential in the chain complex K .

By Proposition 2.24, x survives to the second page if and only if $d_p \circ x$ is null-homotopic, i.e., x is a cycle:

$$d_p \circ x = 0 \quad \text{in} \quad \pi_{n-p} K_{p-1}. \quad (20)$$

Similarly, x survives to the r -th page if it extends to a map of chain complexes over r steps, corresponding to being a cycle up to r -th order.

Let F be a filtered object in $\mathcal{C}$. Recall from Definition 2.14 that we have the spectral sequence $\{E_r^{n,p}, d_r^{n,p}\}$ associated with F . To understand the differentials $d_r^{n,p}$, we consider the following commutative diagram arising from the connecting morphisms in the long exact sequences of homotopy groups:

$$\begin{array}{ccccc} \pi_n(F_p/F_{p-r}) & \longrightarrow & \pi_n(F_p/F_{p-1}) & \longrightarrow & \pi_n(F_{p+r-1}/F_{p-1}) \\ \delta_{p-r} \downarrow & & & & \downarrow \delta_{p-1} \\ \pi_{n-1}(F_{p-r}/F_{p-2r}) & \longrightarrow & \pi_{n-1}(F_{p-r}/F_{p-r-1}) & \longrightarrow & \pi_{n-1}(F_{p-1}/F_{p-r-1}), \end{array} \quad (21)$$

where the vertical maps δ_{p-r} and δ_{p-1} are the connecting homomorphisms from the long exact sequences associated with the cofiber sequences in the filtration.

Let $x \in E_r^{n,p}$ be an element that survives to the r -th page, and let $x' \in \pi_n(F_p/F_{p-r})$ be a lift of x . By the commutativity of Diagram Equation (21), the image of $\delta_{p-r}(x')$ in $\pi_{n-1}(F_{p-1}/F_{p-r-1})$ depends only on x and not on the specific choice of the lift x' . This property allows us to define the differential $d_r^{n,p}$.

Definition 2.29. Let F be a filtered spectrum with spectral sequence $E_r^{n,p}$ and let $x \in E_r^{n,p}$. Let $x' \in \pi_n(F_p/F_{p-r})$ be a lift of x . We define the differential

$$d_r^{n,p}(x) \in E_r^{n-1,p-r}$$

to be the image of $\delta_{p-r}(x')$ in $E_r^{n-1,p-r}$ under the natural map induced by the composition

$$\pi_{n-1}(F_{p-r}/F_{p-2r}) \rightarrow E_r^{n-1,p-r}.$$

Diagrammatically, this is represented as:

$$\begin{array}{ccc} x' & \longrightarrow & x \\ \delta_{p-r} \downarrow & & \\ \delta_{p-r}(x') & \longmapsto & d_r^{n,p}(x). \end{array} \quad (22)$$

The definition ensures that $d_r^{n,p}(x)$ is well-defined, independent of the choice of lift x' .

Chain-like description of the differential

Similar to the description of the pages, as maps between chain complexes, we can unwind Lurie's description for filtered spectrum, to get a chain complex description of the differential.

To facilitate the proof of [Proposition 2.32](#), we first introduce the following definition and lemma.

Definition 2.30. We denote the "shift to the left" functor by

$$T: \text{Ch}(\mathbf{Sp}) \rightarrow \text{Ch}(\mathbf{Sp})$$

which is given by precomposing with

$$-1: \mathbb{Z} \rightarrow \mathbb{Z}, \quad n \mapsto n - 1$$

Lemma 2.31. Let $L \in \text{Ch}_{(\infty, n]}(\mathbf{Sp})$ and $K \in \text{Ch}_{[n, -\infty)}(\mathbf{Sp})$ be two chain complexes. Then

$$\text{Map}(\Sigma^{-1}TL, K) \simeq \text{Map}(L, K)$$

Proof. First, note that $\widehat{\text{Fil}}(\mathbf{Sp})$ is generated (under colimits) by $\mathfrak{J}_p$. Passing to $\text{Ch}(\mathbf{Sp})$, we get that $\text{Ch}(\mathbf{Sp})$ is generated by $\mathcal{I}\mathfrak{J}_p = \mathbb{S}_p^{-p}$. Using this observation, it suffices to show that:

$$\text{Map}(\mathbb{S}_p^{-p}, K) \simeq \text{Map}(\Sigma^{-1}T\mathbb{S}_p^{-p}, K) \simeq \text{Map}(\mathbb{S}_{p+1}^{-(p+1)}, K)$$

and by taking $\mathcal{I}$, our goal is to show:

$$\text{Map}(\mathfrak{J}_p, \mathcal{I}K) \simeq \text{Map}(\mathfrak{J}_{p+1}, \mathcal{I}K)$$

which is equivalent to

$$(\mathcal{I}K)_p \simeq (\mathcal{I}K)_{p+1} \quad (23)$$

Note that the assumption on L translates to $p \geq n$.

Let $F = \mathcal{I}K$, where $K \in \text{Ch}_{[n, -\infty)}(\mathbf{Sp})$. Then for $q \geq n + 1$ we have

$$0 = K_q = (\mathcal{A}F)_q = \Sigma^{-q}F_q/F_{q-1}$$

which shows that [Eq. \(23\)](#) holds. □

Example. To get the direction right, note that $T\mathbb{S}_0 = \mathbb{S}_1$ and get that:

$$\text{Map}(\mathbb{S}_1, \mathbb{S}_0) \simeq \text{Map}(\Sigma^{-1}T\mathbb{S}_0^1, \mathbb{S}_0) \simeq \text{Map}(\mathbb{S}_0^1, \mathbb{S}_0).$$

So shifting the source to the right shifts the homotopical degree of the source one up.

Proposition 2.32. *Let $K \in \text{Ch}(\mathbf{Sp})$ be a chain complex in Spectra, let $F = \mathcal{I}K$ be its associated filtered spectrum, and let $E_r^{n,p}$ be the spectral sequence of F . Let $x \in \mathcal{E}_r^{n,p}$ be an element surviving to the r -th page (see [Definition 2.17](#)). Then the differential $\partial_r(x)$ is given by the composition of chain maps:*

$$\begin{array}{ccccc} \mathbb{S}^{n-p} & \longrightarrow & \cdots & \longrightarrow & 0 \\ \downarrow x & & \downarrow & & \downarrow \\ K_p & \longrightarrow & \cdots & \longrightarrow & K_{p-r+1} \\ \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & \cdots & \longrightarrow & K_{p-r} \end{array}$$

viewed as a map $\mathbb{S}^{n-p} \rightarrow K_{p-r}$, where the first map is an extension of x (see [Proposition 2.24](#)), and the second is induced by the structure of the chain complex K .

Proof verification needed. G2 — gap in the identification asserted above. The coend of [Proposition 5.5](#) gives an explicit map $\Sigma^{n-p+r-1}\mathbb{S} \rightarrow K_{p-r}$, whereas the filtered definition uses a connecting morphism. Compare these as coherent maps under the filtered-object/chain equivalence, with the chosen suspension convention. Lurie [\[19, Construction 1.2.2.6 and Definition 1.2.2.9\]](#) gives the filtered construction, but does not identify this particular endpoint coend with it. The related collapse comparison in [\[11, Proposition 2.24\]](#) does not supply that identification either. No exact reference for the remaining comparison has been located; the geometric differential theorem depends on it.

Proof. We begin with Lurie’s description, [Eq. \(10\)](#). The differential is given by the map:

$$\pi_n F_p / F_{p-r} \rightarrow \pi_{n-1} F_{p-r} / F_{p-2r} \quad (24)$$

hence ∂_r is induced by post-composing with the connecting morphism:

$$F_p / F_{p-r} \rightarrow \Sigma F_{p-r} / F_{p-2r}$$

which comes from the cofiber sequence:

$$F_{p-r} / F_{p-2r} \rightarrow F_p / F_{p-2r} \rightarrow F_p / F_{p-r}.$$

As we saw in the proof of [Proposition 2.24](#), we can use [Lemma 2.23](#) to see that the map $F_p / F_{p-2r} \rightarrow F_p / F_{p-r}$ is representable as

$$j^*: \text{Map}(\mathbb{S}_p^{-p}, K_{[p, p-2r+1]}) \xrightarrow{j^\circ} \text{Map}(\mathbb{S}_p^{-p}, K_{[p, p-r+1]})$$

which is given by postcomposing with restriction and padding. By taking the cofiber of j , we get the cofiber of j^* . The cofiber is calculated pointwise: for $q \in [p-r, p-2r+1]$, the map j^* is $K_q \rightarrow 0$, and for $q \notin [p-r, p-2r+1]$, the map is the identity. We conclude that the cofiber map is of the form:

$$c: K_{[p, p-r+1]} \rightarrow \text{cof}(j^*) \simeq \Sigma K_{[p-r, p-2r+1]}.$$

Hence, the differential is represented by post-composing:

$$c^*: \text{Map}(\mathbb{S}_p^{-p}, K_{[p, p-r+1]}) \xrightarrow{c^\circ} \text{Map}(\mathbb{S}_p^{-p}, \Sigma K_{[p-r, p-2r+1]}).$$

Note that $\mathbb{S}_p^{-p} \in \text{Ch}_{(\infty,p]}(\mathbf{Sp})$ and the other two chains are elements in $\text{Ch}_{[p,-\infty)}(\mathbf{Sp})$. So we can use [Lemma 2.31](#) on c and find that c^* is equivalent to post-composition with

$$K_{[p,p-r+1]} \rightarrow T^{-1}K_{[p-r,p-2r+1]}$$

which is the visualization in the proposition. Note that the diagram (which has K_{p-r} as a target) is a visualization of $\partial_r(x) \in E_1^{n-1,p-r}$, and the "non-truncated" version (with target $K_{[p-r,p-2r+1]}$) is the evidence that we can lift $\partial_r(x)$ to $\mathcal{E}_r^{n-1,p-r}$.

Let $K = \mathcal{AF}$, we get a cofiber sequence of chain maps:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & \cdots & \longrightarrow & 0 & \longrightarrow & K_{p-r} & \longrightarrow & \cdots & \longrightarrow & K_{p-2r+1} \\ \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\ K_p & \longrightarrow & \cdots & \longrightarrow & K_{p-r+1} & \longrightarrow & K_{p-r} & \longrightarrow & \cdots & \longrightarrow & K_{p-2r+1} \\ \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\ K_p & \longrightarrow & \cdots & \longrightarrow & K_{p-r+1} & \longrightarrow & 0 & \longrightarrow & \cdots & \longrightarrow & 0 \end{array}$$

Colimits in the functor category of chains are computed pointwise, hence the connecting morphism is given by

$$\begin{array}{ccccccccc} K_p & \longrightarrow & \cdots & \longrightarrow & K_{p-r+1} & \longrightarrow & 0 & \longrightarrow & \cdots & \longrightarrow & 0 \\ \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & \cdots & \longrightarrow & 0 & \longrightarrow & \Sigma K_{p-r} & \longrightarrow & \cdots & \longrightarrow & \Sigma K_{p-2r+1} \end{array} \quad (25)$$

We denote the restriction of chains:

$$\begin{aligned} K_{[p,p-r+1]} &:= K_p \rightarrow \cdots \rightarrow K_{p-r+1} \rightarrow \overbrace{0 \rightarrow \cdots \rightarrow 0}^{r-1} \\ K_{[p-r,p-2r+1]} &:= \overbrace{0 \rightarrow \cdots \rightarrow 0}^{r-1} \rightarrow K_{p-r} \rightarrow \cdots \rightarrow K_{p-2r+1} \end{aligned}$$

as objects in $\text{Ch}_{[p,p-2r+2]}(\mathbf{Sp})$. The diagram [Eq. \(25\)](#) is equivalent to the diagram:

$$\begin{array}{ccc} K_{[p,p-r+1]} & \longrightarrow & 0 \\ \downarrow & & \downarrow \\ 0 & \longrightarrow & \Sigma K_{[p-r,p-2r+1]} \end{array}$$

Proof verification needed. G2 — the following inference is not justified by stability alone. A map $A \rightarrow \Sigma B$ is a map $\Sigma^{-1}A \rightarrow B$, not a map $A \rightarrow B$. The displayed diagram below instead shifts the index of the lower chain. Specify the natural equivalence using [Lemma 2.31](#), and check that it takes this connecting morphism to the endpoint map supplied by the structure maps of K , including higher coherences and the boundary-sphere sign. Agreement of the objects, or the one-degree calculation alone, does not prove this.

Using the stability of $\mathbf{Sp}$, we get that $\text{Ch}_{[p,p-2r+2]}(\mathbf{Sp})$ is stable, and the last diagram is equivalent to the map of chains: $K_{[p,p-r+1]} \rightarrow K_{[p-r,p-2r+1]}$ which can be drawn as:

$$\begin{array}{ccccccc} K_p & \longrightarrow & \dots & \longrightarrow & K_{p-r+1} & \longrightarrow & \dots & \longrightarrow & 0 \\ \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & \dots & \longrightarrow & K_{p-r} & \longrightarrow & \dots & \longrightarrow & K_{p-2r+1} \end{array}$$

Hence, pre-composition with a lift of x to a map $x' : \mathbb{S}_p^{n-p} \rightarrow K_{[p,p-r+1]}$ gives the diagram:

$$\begin{array}{ccccccc} \mathbb{S}^{n-p} & \longrightarrow & \dots & \longrightarrow & 0 & \longrightarrow & \dots & \longrightarrow & 0 \\ \downarrow x & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\ K_p & \longrightarrow & \dots & \longrightarrow & K_{p-r+1} & \longrightarrow & \dots & \longrightarrow & 0 \\ \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & \dots & \longrightarrow & K_{p-r} & \longrightarrow & \dots & \longrightarrow & K_{p-2r+1} \end{array} \quad (26)$$

and the truncated version:

$$\begin{array}{ccccc} \mathbb{S}^{n-p} & \longrightarrow & \dots & \longrightarrow & 0 \\ \downarrow & & \downarrow & & \downarrow \\ K_p & \longrightarrow & \dots & \longrightarrow & K_{p-r+1} \\ \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & \dots & \longrightarrow & K_{p-r} \end{array}$$

The last diagram is $\partial_r(x) \in E_1^{n-1,p-r}$, and the previous diagram Eq. (26) is the evidence that we can lift $\partial_r(x)$ to $\mathcal{E}_r^{n-1,p-r}$. $\square$

Theorem B. Let $K \in \text{Ch}(\mathbf{Sp})$ be a coherent chain complex in Spectra. Then there is a spectral sequence:

$$E_1^{n,p} = \pi_{n-p} K_p,$$

which conditionally converges to

$$E_r^{n,p} \implies \pi_n \text{colim}_p (\mathcal{I}K)_p,$$

Moreover, let $x : \mathbb{S}^{n-p} \rightarrow K_p$ represent an element of $E_1^{n,p}$. Then x survives to $E_{r+1}^{n,p}$ if and only if there exists a chain complex map:

$$\begin{array}{ccccccc} \mathbb{S}^{n-p} & \longrightarrow & 0 & \longrightarrow & \dots & \longrightarrow & 0 \\ \downarrow x & & \downarrow & & \downarrow & & \downarrow \\ K_p & \xrightarrow{d_p} & K_{p-1} & \xrightarrow{d_{p-1}} & \dots & \xrightarrow{d_{p-r}} & K_{p-r} \end{array} \quad (2)$$

Here, the top row is concentrated in degree p , and the bottom row is the restriction of K from degree p down to $p - r$. The differential $\partial_{r+1}(x)$ is given by the composition of chain maps:

$$\begin{array}{ccccc}
 \mathbb{S}^{n-p} & \longrightarrow & \dots & \longrightarrow & 0 \\
 \downarrow x & & \downarrow & & \downarrow \\
 K_p & \longrightarrow & \dots & \longrightarrow & K_{p-r} \\
 \downarrow & & \downarrow & & \downarrow \\
 0 & \longrightarrow & \dots & \longrightarrow & K_{p-r-1}
 \end{array}$$

The first map is precisely [Eq. \(2\)](#) and the second is induced by the structure of the chain complex K .

Lemma 2.33. *Let $K \in \text{Ch}(\mathbf{Sp})$, let $F = \mathcal{I}K$, and let $r \geq 1$. Suppose that $x: \mathbb{S}^{n-p} \rightarrow K_p$ represents an element of $\mathcal{E}_r^{n,p} \subseteq E_1^{n,p}$, and write $[x]_r \in E_r^{n,p}$ for its class. For an extension $\tilde{x}: \mathbb{S}_p^{n-p} \rightarrow K_{[p,p-r+1]}$, write $\partial_r(\tilde{x}) \in \mathcal{E}_r^{n-1,p-r} \subseteq E_1^{n-1,p-r}$ for the representative of $d_r^{n,p}([x]_r)$ given by [Proposition 2.32](#). The following conditions are equivalent:*

1. $d_r^{n,p}([x]_r) = 0$ in $E_r^{n-1,p-r}$;
2. there is an extension $\tilde{x}$ of x for which $\partial_r(\tilde{x}) = 0$ in $E_1^{n-1,p-r}$;
3. x extends to a map of coherent chain complexes $\mathbb{S}_p^{n-p} \rightarrow K_{[p,p-r]}$.

Proof. By [Lemma 2.23](#) and [Proposition 2.24](#), an extension $\tilde{x}$ corresponds to a lift $\alpha \in \pi_n(F_p/F_{p-r})$ of x . Its obstruction $u = \partial_r(\tilde{x})$ is the image of α under the connecting homomorphism of the cofiber sequence

$$F_{p-r}/F_{p-r-1} \longrightarrow F_p/F_{p-r-1} \longrightarrow F_p/F_{p-r}.$$

Thus $u \in \pi_{n-1}(F_{p-r}/F_{p-r-1}) = E_1^{n-1,p-r}$. By [Definition 2.14](#) and [Proposition 2.32](#), the image of u in $\pi_{n-1}(F_{p-1}/F_{p-r-1})$ is $d_r^{n,p}([x]_r)$, viewed in this group via the inclusion in [Eq. \(8\)](#).

Suppose this image vanishes. Exactness for the cofiber sequence

$$F_{p-r}/F_{p-r-1} \longrightarrow F_{p-1}/F_{p-r-1} \longrightarrow F_{p-1}/F_{p-r}$$

identifies the target boundary subgroup, in the usual notation, as

$$\begin{aligned}
 B_r^{n-1,p-r} &= \text{im}(\pi_n(F_{p-1}/F_{p-r}) \xrightarrow{\partial} E_1^{n-1,p-r}) \\
 &= \ker(E_1^{n-1,p-r} \rightarrow \pi_{n-1}(F_{p-1}/F_{p-r-1})).
 \end{aligned}$$

It therefore gives $\beta \in \pi_n(F_{p-1}/F_{p-r})$ whose boundary is u . Let j be the map from $\pi_n(F_{p-1}/F_{p-r})$ to $\pi_n(F_p/F_{p-r})$ induced by $F_{p-1} \rightarrow F_p$, and set $\alpha' = \alpha - j(\beta)$. The image of α' in $\pi_n(F_p/F_{p-1})$ is still x . By naturality of the connecting homomorphisms, its obstruction is $u - \partial(j(\beta)) = u - u = 0$. Hence α' gives the extension in (2). Conversely, (2) immediately implies (1).

Exactness for the first cofiber sequence shows that an extension has zero obstruction precisely when its class α' lifts to $\pi_n(F_p/F_{p-r-1})$. By [Proposition 2.24](#), this is precisely an extension of x to $K_{[p,p-r]}$, proving (2) $\Leftrightarrow$ (3). In the same notation, $\mathcal{E}_r$ is Z_r . Exactness of $F_{p-1}/F_{p-r} \rightarrow F_p/F_{p-r} \rightarrow F_p/F_{p-1}$ shows that any two extensions of x differ by some $j(\beta)$. Thus varying the extension changes its raw obstruction by exactly the subgroup B_r displayed above. $\square$

3 Manifolds with Corners and Framed Flow Categories

This section lays the geometric and categorical groundwork for our main constructions. We begin by introducing a diagrammatic framework. The fundamental geometric objects in this setting are manifolds with corners, which generalize smooth manifolds (see [15] for a relevant treatment). These concepts are then used to define flow categories [9], where the morphism spaces are manifolds with corners and composition is given by embedding into the corner strata. The subsequent subsections develop the precise definitions required for this approach.

3.1 Diagrammatic framework

Let A be a set and $\mathcal{C}$ a category. An $\langle A \rangle$ diagram in $\mathcal{C}$ is a functor:

$$F: [\tilde{1}]^A \rightarrow \mathcal{C}.$$

In the case where $C \in \mathcal{C}$ have a name (e.g. space, manifold, group), we call such functor an $\langle A \rangle$ space, etc.

Definition 3.1. Let A be a finite set and $a \in A$ then we define its **A -boundary** to be the restriction along the inclusion

$$i: \text{Fun}(A \setminus \{a\}, [\tilde{1}]) \rightarrow \text{Fun}(A, [\tilde{1}])$$

where i is given by taking the full subcategory spanned by functors with $F(a) = 1$. We get a map:

$$\partial_a: \text{Fun}([\tilde{1}]^A, \mathcal{C}) \rightarrow \text{Fun}([\tilde{1}]^{A \setminus \{a\}}, \mathcal{C})$$

An $\langle A \rangle$ -**map** is a natural transformation between $\langle A \rangle$ -diagrams.

Remark 3.2. Notice that

$$\text{Fun}(1, \text{Fun}([\tilde{1}]^A, \mathcal{C})) \simeq \text{Fun}([\tilde{1}]^A, \text{Fun}(1, \mathcal{C}))$$

hence any $\langle A \rangle$ -map is an $\langle A \rangle$ diagram in the arrow category $\text{Fun}(1, \mathcal{C})$, and we can use $\langle A \rangle$ diagrams notations, like

$$\partial_a \mu$$

Definition 3.3. Let $\mathcal{C}$ be a monoidal category. For $i \in \{1, 2\}$, let A_i be finite sets and F_i an $\langle A_i \rangle$ diagram. Then there is a tensor product

$$F_1 \otimes F_2(B_1 \sqcup B_2) := F_1(B_1) \otimes F_2(B_2).$$

which is an $\langle A_1 \sqcup A_2 \rangle$ diagram in $\mathcal{C}$. Given two morphisms $\mu_i: F_i \rightarrow G_i$ we define

$$(\mu_1 \otimes \mu_2)(B_1 \sqcup B_2) := \mu_1(B_1) \otimes \mu_2(B_2)$$

which is a map of $\langle A_1 \sqcup A_2 \rangle$ diagrams.

Taking $\mathcal{C} = \mathbf{Top}$ and let A be a finite set, here are few useful constructions that we will use later. We remind the reader that the one point compactification is a monoidal functor from locally compact Hausdorff spaces with proper maps with disjoint union (resp. Cartesian product) to pointed spaces with wedge sum (resp. smash product). Moreover it is a contravariant functor from the category of topological spaces with open inclusions, also known as the Pontrjagin-Thom collapse map. Explicitly, given $f : X \rightarrow Y$ and map continuous map between topological spaces the collapse map is the map $Y_+ \rightarrow X_+$

$$y \mapsto \begin{cases} x & y = f(x) \\ \infty_X & \text{otherwise} \end{cases}$$

Let X be $\langle A \rangle$ diagram in $\mathbf{Top}$ such that all the maps are proper. We have an $\langle A \rangle$ diagram, X_+ , given by

$$X_+(B) := (X(B))_+$$

which arising from the functoriality.

Definition 3.4. Let $f : X \rightarrow Y$ be a map of $\langle A \rangle$ diagrams in $\mathbf{Top}$ such that $f(B)$ is open inclusion, then the functoriality of $(-)_+$ yields the **collapse map**

$$f_+ : Y_+ \rightarrow X_+$$

Definition 3.5. Let A be a finite set and M, N two $\langle A \rangle$ manifolds, we say that a map $e : M \rightarrow N$ is **neat embedding** if

1. For each $B \subseteq A$, the map $e(B) : M(B) \rightarrow N(B)$ is an embedding.
2. The intersections $M(B) \cap N(C)$ are transverse for all $B \subseteq C$.
3. The preimage of the faces satisfies $e^{-1}(\partial_a N) = \partial_a M$ for all $a \in A$.

For $i \in \{1, 2\}$ let A_i be finite sets and let $e_i : M_i \rightarrow N_i$ be an two neat embeddings of $\langle A \rangle_i$ diagrams of manifolds. Taking the monoidal structure $(\mathbf{Top}, \times)$ we can use [Definition 3.3](#) and get a map

$$e_1 \times e_2 : M_1 \times M_2 \rightarrow N_1 \times N_2$$

which is a map of $\langle A_1 \sqcup A_2 \rangle$ diagrams.

Moreover, we can define the **normal bundle**, ν_{e_i} , by

$$(\nu_{e_i})(B) := \nu_{e_i}(B)$$

which are $\langle A \rangle$ diagrams in $\mathbf{Top}$ of the total spaces of the bundles. We can do the same for the map $e_1 \times e_2$ and the fact that the total space of normal bundle commute with Cartesian product of the base, yields:

$$\nu_{e_1} \times \nu_{e_2} = \nu_{e_1 \times e_2}$$

as $\langle A_1 \sqcup A_2 \rangle$ diagrams.

Lastly, we have open inclusions of diagrams:

$$\nu_{e_i} \rightarrow N_i, \quad \nu_{e_1} \times \nu_{e_2} \rightarrow N_1 \times N_2$$

hence we can compactify and get maps

$$(N_i)_+ \rightarrow (\nu e_i)_+, \quad (N_1 \times N_2)_+ \rightarrow (\nu_{e_1} \times \nu_{e_2})_+$$

and the monoidality of $(-)_+$ give rise to a map of $\langle A_1 \sqcup A_2 \rangle$ diagrams:

$$(N_1)_+ \wedge (N_2)_+ \rightarrow (\nu_{e_1})_+ \wedge (\nu_{e_2})_+.$$

Definition 3.6. Given an $\langle A \rangle$ vector bundle over an $\langle A \rangle$ manifold, i.e. an $\langle A \rangle$ map $\pi : E \rightarrow M$, such that $\pi(B)$ is vector bundle for $A \subseteq A$, a **trivialization** of E is a map of $\langle A \rangle$ -spaces

$$\varphi : E \simeq M \times \mathbb{R}^N$$

for some $N \in \mathbb{N}$, where $\mathbb{R}^N$ is the $\langle \emptyset \rangle$ diagram on $\mathbb{R}^N$. This map amounts to family of bundle maps

$$\varphi(B) : E(B) \simeq M(B) \times \mathbb{R}^N.$$

We can project to the second coordinate of the RHS and get a map

$$E \rightarrow \underline{\mathbb{R}^N},$$

where the RHS is the constant functor on $\mathbb{R}^N$. Applying $(-)_+$ resulting an $\langle A \rangle$ -map

$$E_+ \rightarrow \underline{(\mathbb{R}^N)_+} \simeq \underline{S^N}. \quad (27)$$

Our main application is the case where we have a neat embedding of $\langle A \rangle$ manifolds

$$e : M \hookrightarrow \mathbb{R}^N \times \mathbb{R}_{0 \leq}^A$$

and a framing of the normal bundle

$$\varphi : \nu_e \simeq M \times \mathbb{R}^{N+|A|-n}.$$

We construct the **Thom map**:

$$f_{e,\varphi} : (\mathbb{R}^N \times \mathbb{R}_{0 \leq}^A)_+ \rightarrow (\nu_e)_+ \rightarrow (\mathbb{R}^{N+|A|-n})_+$$

where the first map is the collapse map of e and the second map is the projection of φ to the second coordinate, and applying $(-)_+$.

If $|A| = n$ we can write

$$f : S^N \wedge (\mathbb{R}_{0 \leq}^A)_+ \rightarrow (\nu_e)_+ \rightarrow S^N. \quad (28)$$

3.2 Manifolds with corners

Let $\mathbb{R}_{0 \leq}$ denote the half-line:

$$\mathbb{R}_{0 \leq} := [0, \infty).$$

Definition 3.7. An n -dimensional **manifold with corners** is a topological space M together with an atlas, modeled on open subsets of $\mathbb{R}_{0 \leq}^n$ with smooth transition functions. We define $c(m)$ as the number of coordinates equal to zero in a chart around m . A **connected face** of M is a closures of connected components of:

$$\{m \in M \mid c(m) = 1\}.$$

A **face** is a union of connected faces.

An n -dimensional **manifolds with faces** is an n -dimensional manifold with corners such that every $m \in M$ belong to $c(m)$ connected faces of M .

Note that a face of an n -dimensional manifolds with faces is an $(n - 1)$ -dimensional manifolds with faces.

Definition 3.8. Let A be a finite set. An $\langle A \rangle$ -**manifold** is a manifold with faces M , together with a collection of faces $\{\partial_a M\}_{a \in A}$ satisfying the following conditions:

1. The union of all faces covers the boundary of M :

$$\bigcup_{a \in A} \partial_a M = \partial M.$$

2. For $a \neq b$, the intersection $\partial_a M \cap \partial_b M$ is a face of both $\partial_a M$ and $\partial_b M$.

If $A = \{1, \dots, k\}$, we refer to M as a $\langle k \rangle$ -**manifold**.

Note. Using the definitions of $\langle A \rangle$ -manifold, we get that $\partial_a M$ inherit a structure of an $\langle A \setminus a \rangle$ -manifold.

Example 3.9. The space $\mathbb{R}_{0 \leq}^A$ admits an $\langle A \rangle$ -manifold structure by defining the faces:

$$\partial_a \mathbb{R}_{0 \leq}^A = \mathbb{R}_{0 \leq}^{A \setminus \{a\}} = \{x \in \mathbb{R}_{0 \leq}^A \mid x_a = 0\}.$$

More generally, $\mathbb{R}^N \times \mathbb{R}_{0 \leq}^A$ can be given an $(N + |A|)$ -dimensional $\langle A \rangle$ -manifold structure by setting:

$$\partial_a(\mathbb{R}^N \times \mathbb{R}_{0 \leq}^A) = \mathbb{R}^N \times \partial_a \mathbb{R}_{0 \leq}^A.$$

Example 3.10. Consider a triangle in the plane. It is a 2-dimensional manifold with corners, which admits a $\langle 3 \rangle$ -manifold structure by taking the three edges as the faces $\partial_a M$. This example illustrates that not all k -manifolds are necessarily k -dimensional. However, in this paper, we focus on manifolds where the dimension equals the number of faces, i.e., $n = k$.

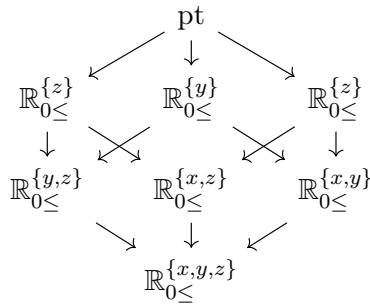
Construction 3.11. Let M be an n -dimensional $\langle A \rangle$ -manifold. It induces an $\langle A \rangle$ -diagram in the category of manifolds with embeddings by defining:

$$M(B) = \bigcap_{b \in B} \partial_b M,$$

for each subset $B \subseteq A$. In particular, $M(\emptyset) = M$, and $M(B)$ is an $(n - |B|)$ -dimensional $\langle B \rangle$ -manifold.

Remark. For simplicity, we sometimes refer to an $\langle A \rangle$ -manifold by its associated $\langle A \rangle$ -diagram.

Example 3.12. The manifold $\mathbb{R}_{0 \leq}^3$ induces the following $\langle 3 \rangle$ -diagram:



The top vertex represents the origin. The second row consists of the coordinate axes $\mathbb{R}_{0\leq}^{\{x\}}$, $\mathbb{R}_{0\leq}^{\{y\}}$, and $\mathbb{R}_{0\leq}^{\{z\}}$. The third row contains the coordinate planes, which correspond to the faces $\partial_a \mathbb{R}_{0\leq}^3$. The bottom vertex is the entire manifold $\mathbb{R}_{0\leq}^3$.

Construction 3.13. For $i \in \{1, 2\}$, let M_i be an n -dimensional $\langle A \rangle_i$ -manifold, then the Cartesian product $M_1 \times M_2$ admits an $\langle A_1 \sqcup A_2 \rangle$ -manifold structure by

$$\partial_a(M_1 \times M_2) := \begin{cases} (\partial_a M_1) \times M_2 & a \in A_1 \\ M_1 \times (\partial_a M_2) & a \in A_2 \end{cases}$$

Note that this construction coincides with [Definition 3.3](#).

Definition 3.14. A neat embedding of $\langle A \rangle$ -manifolds with corners is a map of topological spaces $e : M \rightarrow N$ such that the induces map of $\langle A \rangle$ -diagrams is neat embedding.

Definition 3.15. Let M be an n dimensional $\langle A \rangle$ manifold, $e : M \hookrightarrow \mathbb{R}^N \times \mathbb{R}_{0\leq}^A$, a neat embedding. Note that the normal bundle is an $\langle A \rangle$ vector bundle over M , of rank $N + |A| - n$. A **normal framing** is a trivialization (see [Definition 3.6](#)) of ν_e

$$\varphi : \nu_e \simeq M \times \mathbb{R}^{N+|A|-n}.$$

We denote by

$$Th(M, e) := (\nu_e)_+$$

the **Thom space** of M and e , which is an $\langle A \rangle$ space.

If the choice of neat embedding is clear, we will omit e and write $Th(M)$.

Remark. Let n be the dimension of M , each normal bundle $\nu_e M(B)$ is of rank $N + |A| - n$.

Similar to [Construction 3.13](#), for $i \in \{1, 2\}$, let M_i be an embedded manifold with normal framing φ_i . Since the normal bundle of a Cartesian product is the direct sum of the individual normal bundles, the Cartesian product $M_1 \times M_2$ admits a normal framing. We denote the resulting framing by $\varphi_1 \oplus \varphi_2$.

3.3 Framed Flow Categories

We now introduce framed flow categories, first developed by Cohen, Jones, and Segal [9] to axiomatize the geometric data arising from Morse-Smale functions and certain Floer-theoretic setups. A flow category is a graded category whose morphism spaces are manifolds with corners, with dimensions determined by the grading. Composition is realized geometrically by identifying the boundary strata of one morphism manifold with the products of others.

A *framed* flow category further equips these morphism manifolds with a stable framing that must be compatible with the composition law. This coherent structure is what allows for the construction of a stable homotopy type. Our treatment, which relies on neat embeddings of these manifolds into Euclidean space, follows the perspective used in modern applications like [17].

Definition 3.16. Let $\mathcal{C}$ be a category with a **grading function**:

$$|-| : \text{ob}(\mathcal{C}) \rightarrow \mathbb{Z}.$$

For each pair of objects $x, y \in \mathcal{C}$, define the linearly ordered set:

$$J_{y,x} := \{i \in \mathbb{Z} \mid |y| > i > |x|\}. \quad (29)$$

Definition 3.17. A **flow category** is a topologically enriched category $\mathcal{C}$ with a discrete set of objects and a grading function

$$|-| : \text{ob}(\mathcal{C}) \rightarrow \mathbb{Z}$$

Moreover, for each pair $x, y \in \mathcal{C}$ the morphism space $\text{Hom}_{\mathcal{C}}(y, x)$ is endowed with a structure of an $|J_{y,x}|$ -dimensional $\langle J_{y,x} \rangle$ -manifold, also denoted by $M_{y,x}$, such that the corner structure satisfies:

$$\partial_i M_{z,x} = \bigsqcup_{y \in \text{ob}(\mathcal{C}), |y|=i} M_{z,y} \times M_{y,x}, \quad (30)$$

for all $i \in J_{z,x}$. The composition is given by the inclusion of the boundary:

$$M_{z,y} \times M_{y,x} \hookrightarrow \partial_i M_{z,x} \hookrightarrow M_{z,x}.$$

Notation 3.18. Let $\mathcal{C}$ be a flow category and $z, y, x \in \mathcal{C}$, we will write $\partial_y M_{z,x}$ to denote $M_{z,y} \times M_{y,x}$ as connected component of the the face $\partial_i M_{z,x}$.

Notation 3.19. Let $\mathcal{C}$ be a flow category and $j, i \in \mathbb{Z}$ we denote

$$M_{i,j} := \bigsqcup_{\substack{y \in \mathcal{C}, |y|=j \\ x \in \mathcal{C}, |x|=i}} M_{y,x}$$

Example 3.20. Consider a framed flow category $\mathcal{C}$ with object set $\{z, y, x, w\}$ and grading $|z| = 4, \dots, |w| = 1$. The morphism spaces are:

- Zero-dimensional manifolds $M_{z,y}, M_{y,x}, M_{x,w}$, which are discrete points (possibly with signs).
- One-dimensional framed manifolds $M_{z,x}$ and $M_{y,w}$, serving as null-bordisms:

$$\partial_3 M_{z,x} = M_{z,y} \times M_{y,x}, \quad \partial_2 M_{y,w} = M_{y,x} \times M_{x,w}.$$

- A two-dimensional manifold $M_{z,w}$ with faces:

$$\partial_3 M_{z,w} = M_{z,y} \times M_{y,w}, \quad \partial_2 M_{z,w} = M_{z,x} \times M_{x,w},$$

and corners given by $M_{z,y} \times M_{y,x} \times M_{x,w}$.

Definition 3.21. A *neat embedding* of a flow category $\mathcal{C}$ consists of a choice of integer N and a family of neat embeddings of manifolds with corners,

$$e_{y,x} : M_{y,x} \hookrightarrow \mathbb{R}^{N(|y|-|x|)} \times \mathbb{R}_{0 \leq}^{J_{y,x}},$$

for each pair of objects y, x , such that the following diagram of $\langle J_{z,x} \setminus \{|y|\} \rangle$ -spaces, commutes

$$\begin{array}{ccc} M_{z,y} \times M_{y,x} & \xrightarrow{e_{z,y} \times e_{y,x}} & \mathbb{R}^{N(|z|-|y|)} \times \mathbb{R}_{0 \leq}^{J_{z,y}} \times \mathbb{R}^{N(|y|-|x|)} \times \mathbb{R}_{0 \leq}^{J_{y,x}} \\ \downarrow & & \downarrow \\ \partial_y M_{z,x} & \xrightarrow{\partial_i e_{z,x}} & \mathbb{R}^{N(|z|-|x|)} \times \partial_i \mathbb{R}_{0 \leq}^{J_{z,x}} \end{array}$$

where $i = |y|$, the bottom map is the restriction of $\partial_i e_{z,x}$ to the y component the top map is [Definition 3.3](#) and the vertical maps are diffeomorphisms.

Note that is a uniform version, in the sense that all the manifolds embeds into a products of $\mathbb{R}^N$. This version arises by stabilizing [15].

Remark 3.22. The fact that any manifold with corners admits a neat embedding implies that any flow category can be embedded, see [15] Lemma 3.16. Moreover, the space of such embeddings becomes arbitrarily connected as N increases.

do we need this? (wait for index bundle change first)

We recall that for two bundles $E_i \rightarrow M_i$, we have the following.

$$Th(E_1 \boxtimes E_2) \simeq Th(E_1) \wedge Th(E_2)$$

where the LHS is the Thom space of the bundle over $M_1 \times M_2$. Using the functoriality of the collapse map (Definition 3.4, Definition 3.4), and the fact above, we get a commutative diagram of $\langle A \rangle$ -spaces

$$\begin{array}{ccc} S^{N(|z|-|x|)} \wedge (\mathbb{R}_{0 \leq}^{J_{z,y}} \times \mathbb{R}_{0 \leq}^{J_{y,x}})_+ & \longrightarrow & Th(M_{z,y}) \wedge Th(M_{y,x}) \xrightarrow{\simeq} Th(M_{z,y} \times M_{y,x}) \\ \downarrow & & \downarrow \\ S^{N(|z|-|x|)} \wedge (\partial_i \mathbb{R}_{0 \leq}^{J_{z,x}})_+ & \longrightarrow & \partial_y Th(M_{z,x}) \end{array} \quad (31)$$

Definition 3.23. Let $\mathcal{C}$ be an embedded flow category. A normal **framing** of $\mathcal{C}$ is a choice of normal framing

$$\varphi_{y,x}: \nu M_{y,x} \simeq M_{y,x} \times \mathbb{R}^{N(|y|-|x|)}$$

for each two objects $x, y \in \mathcal{C}$, with the following coherency

$$\partial_y \varphi_{z,x} = \varphi_{z,y} \oplus \varphi_{y,x}.$$

The compatibility of the neat embedding and the framing implies the commutativity of the diagram:

$$\begin{array}{ccccc} S^{N(|z|-|x|)} \wedge (\mathbb{R}_{0 \leq}^{J_{z,y}} \times \mathbb{R}_{0 \leq}^{J_{y,x}})_+ & \longrightarrow & Th(M_{z,y} \times M_{y,x}) & \xrightarrow{\varphi_{z,y} \oplus \varphi_{y,x}} & \underline{S^{N(|z|-|y|)} \wedge S^{N(|y|-|x|)}} \\ \downarrow & & \simeq \downarrow & & \downarrow \text{id} \\ S^{N(|z|-|x|)} \wedge (\partial_i \mathbb{R}_{0 \leq}^{J_{z,x}})_+ & \longrightarrow & Th(\partial_y M_{z,x}) & \xrightarrow{\partial_y \varphi_{z,x}} & \underline{S^{N(|z|-|x|)}}. \end{array} \quad (32)$$

where the vertical maps are the "Thom" maps; see Eq. (28).

3.4 Flow Modules

Definition 3.24. Let $\mathcal{C}$ be a flow category. A **right flow module of degree n** over $\mathcal{C}$ consists of compact manifolds with faces W_x of dimension $n - |x|$, with $W_x = \emptyset$ if this dimension is negative, and specified face identifications

$$\partial_i W_x = \bigsqcup_{\substack{y \in \mathcal{C} \\ |y|=i}} W_y \times M_{y,x}, \quad |x| < i \leq n.$$

These faces cover the whole boundary of W_x . Their iterated intersections are the products

$$W_{y_k} \times M_{y_k, y_{k-1}} \times \cdots \times M_{y_1, x}$$

for strictly decreasing grading chains $|y_k| > \cdots > |y_1| > |x|$, with the disjoint unions over objects of the specified grades understood. All ways of identifying an iterated face agree with flow composition and the module action, and there are no further boundary faces.

Definition 3.25. Let $\mathcal{C}$ be a flow category. A **left flow module of degree q** over $\mathcal{C}$ consists of compact manifolds with faces V_y of dimension $|y| - q$, empty in negative dimension, and specified face identifications

$$\partial_i V_y = \bigsqcup_{\substack{x \in \mathcal{C} \\ |x|=i}} M_{y,x} \times V_x, \quad q \leq i < |y|.$$

These faces cover the whole boundary of V_y . Their higher intersections are exactly the iterated products of flow manifolds followed by a module manifold; the identifications agree with flow composition on every iterated face.

Definition change to review. The right and left module definitions now require compact manifolds with faces and identify every boundary face with the stated products, including all higher intersections. The previous wording specified compatible boundary embeddings but did not explicitly exclude additional boundary faces. The framed definition below also makes compatibility of the embeddings explicit. These conditions are used to obtain a closed glued manifold; compare to the original definition. The intended dimensions and directions of the actions are unchanged.

Definition 3.26. Let $\mathcal{C}$ be a neatly embedded, normally framed flow category. A **neatly embedded, normally framed right flow module** W of degree n over $\mathcal{C}$ consists of a right flow module structure together with a choice of integer N , neat embeddings of manifolds with corners:

$$e_x : W_x \hookrightarrow \mathbb{R}^{N(n+1-|x|)} \times \mathbb{R}_{0 \leq}^{J_{n+1,x}}$$

which restrict on every face $W_y \times M_{y,x}$ to the product of the module and flow embeddings, under the fixed ambient coordinate identifications. These restrictions agree on all higher faces. The module also has stable normal framings φ_x on νe_x , compatible on every face with the product framing:

$$\partial_y \varphi_x = \varphi_y \oplus \varphi_{y,x}$$

where $\varphi_{y,x}$ is the framing on the morphism manifold $M_{y,x}$ of $\mathcal{C}$.

Similarly, we can define a neatly embedded, normally framed left flow module.

A usefull constrution, which allow us to use the flow categories tools on flow modules is the following:

Definition 3.27. Let W be a right flow module of degree i over a flow category $\mathcal{C}$, with flow manifolds $M_{y,x}$. We define the **Cone Flow Category** $\text{Cone}(W)$ as follows:

- The objects are $\text{ob}(\text{Cone}(W)) = \text{ob}(\mathcal{C}) \sqcup \{z\}$, where z is a new formal object with index $|z| = i + 1$.
- For $x, y \in \mathcal{C}$, the morphism spaces remain identical: $(\text{Cone}(W))_{y,x} = M_{y,x}$.

- For $x \in \mathcal{C}$, the morphism space from z to x is defined as the module manifold:
 $(\text{Cone}(W))_{z,x} = W_x$. (Note that $\dim(W_x) = i - |x| = |z| - |x| - 1$, which satisfies the dimension axiom for flow categories).
- The morphism spaces from x to z are empty.

The corner structure maps

$$\coprod_{y \in \mathcal{C}} (\text{Cone}(W))_{z,y} \times (\text{Cone}(W))_{y,x} \hookrightarrow \partial(\text{Cone}(W))_{z,x}$$

are given exactly by the module structure maps $\coprod_y W_y \times M_{y,x} \hookrightarrow \partial W_x$. If $\mathcal{C}$ and W are normally framed, these framings naturally endow the flow category $\text{Cone}(W)$ with a normal framing.

Similarly, for a left flow module V of degree i over $\mathcal{C}$, its **Cone Flow Category** $\text{Cone}(V)$ is defined analogously by adding a new object z in degree $i - 1$. The morphism spaces from $\mathcal{C}$ to z are given by V and the corner structure maps are determined by the left module structure maps.

Definition 3.28. Let $\mathcal{C}$ be a framed flow category, and let $n, p, r \in \mathbb{Z}$ with $r \geq 0$. Let $W_p = \{W_z\}_{|z|=p}$ be a collection of closed framed manifolds representing an element in:

$$E_1^{n,p} = \bigoplus_{|z|=p} \Omega_{n-p}^{\text{fr}}.$$

We say that the collection W_p **extends to a right flow module of degree n over $\mathcal{C}_{[p,p-r]}$** if there exists a neatly embedded, normally framed right flow module W of degree n over $\mathcal{C}_{[p,p-r]}$ whose restriction to the objects of degree p satisfies $W_z = (W_p)_z$ for all $z \in \mathcal{C}_p$.

Example 3.29. Let $\mathcal{C}$ be a framed flow category with objects $\{y_i\}$ and $\{x_j\}$ concentrated in only two degrees, $|y_i| = q$ and $|x_j| = p$. The morphism space:

$$M = \bigsqcup_{i,j} M_{y_i, x_j}$$

is a closed framed manifold of dimension $q - p - 1$, since there are no intermediate objects with degrees between p and q . In this case, $\mathcal{C}$ defines an element in the framed bordism group $\Omega_{q-p-1}^{\text{fr}}$.

4 Chain complex of Flow Category

****Note for rewriting.**** The $\mathcal{J}$ -model comparison and the endpoint/coend interpretation are now given in this section and [Section 5](#). In the final exposition pass, check that these short model explanations suffice for a reader interested primarily in the geometric differential.

In this section, we establish a connection between chain complexes and framed flow categories by introducing a topologically enriched category $\mathcal{J}$, which serves as a replacement for the category of chain complexes Ch . This idea originates from the work of Cohen, Jones, and Segal [9], and is further developed in the revised version [8].

The category $\mathcal{J}$ records differentials and their coherent nullhomotopies by compactified orthants. We use the version of the Cohen–Jones–Segal construction described in [10, 11]. Its nerve presents the indexing category of coherent chains; the passage to derived morphisms is treated in [Proposition 5.1](#).

Definition 4.1. We define the pointed topologically enriched category $\mathcal{J}$ as follows:

- The objects of $\mathcal{J}$ are the integers together with a distinguished object ∞ :

$$\text{ob}(\mathcal{J}) = \mathbb{Z} \cup \{\infty\}.$$

- The morphism spaces $\text{Hom}_{\mathcal{J}}(j, i)$ are defined for $i, j \in \mathbb{Z} \cup \{\infty\}$ by:

$$\text{Hom}_{\mathcal{J}}(j, i) = \begin{cases} \{\text{id}, \infty\} & \text{if } i = j \in \mathbb{Z}, \\ \{d_i, \infty\} & \text{if } i, j \in \mathbb{Z}, j = i + 1, \\ (\mathbb{R}_{0 \leq}^{J_{j,i}})_+ & \text{if } i, j \in \mathbb{Z}, j > i + 1, \\ \{\infty\} & \text{otherwise,} \end{cases}$$

where X_+ denotes the one-point compactification of a space X , and $\mathbb{R}_{0 \leq}$ is the half-line $[0, \infty)$. The set $J_{j,i}$ is defined by:

$$J_{j,i} = \{\ell \in \mathbb{Z} \mid j > \ell > i\}. \quad (33)$$

The spaces $\text{Hom}_{\mathcal{J}}(j, i)$ are pointed at the point ∞ .

- The maps

$$J_{k,j} \sqcup J_{j,i} \hookrightarrow J_{k,i}$$

induce, by inserting zero in coordinate j , a map

$$\mathbb{R}_{0 \leq}^{J_{k,j}} \times \mathbb{R}_{0 \leq}^{J_{j,i}} \hookrightarrow \mathbb{R}_{0 \leq}^{J_{k,i}}.$$

By taking $(-)_+$ we obtain the composition in $\mathcal{J}$:

$$(\mathbb{R}_{0 \leq}^{J_{k,j}})_+ \wedge (\mathbb{R}_{0 \leq}^{J_{j,i}})_+ \rightarrow (\mathbb{R}_{0 \leq}^{J_{k,i}})_+.$$

Remark 4.2. Intuitively, the morphisms in $\mathcal{J}$ correspond to the differentials and higher compositions in a chain complex. The space $\mathbb{R}_{0 \leq}^{J_{j,i}}$ models the higher nullhomotopies between compositions of differentials.

Definition 4.3. Let C be a pointed topologically enriched category. We define (topological, coherent) chain complexes in C as strict functors $\text{Fun}_+(\mathcal{J}, C)$.

Let $\mathbf{Sp}^O$ denote orthogonal spectra with the ordinary stable model structure [21]. Its localization is $\mathbf{Sp}$; based suspension spectra give a symmetric monoidal left Quillen functor. We first construct the collapse diagram in based spaces on a bounded grading window. This keeps the finite-dimensional collapse maps and their face identities visible before any spectral replacement.

Construction 4.4. Let $\mathcal{C}$ be a flow category with a neat embedding e and normal framing φ . Assume that $\mathcal{C}$ has finitely many objects and bounded index

$$b \geq |-| \geq a.$$

After translating degrees, take $a = 0$. Choose one Euclidean block $\mathbb{R}^N$ for each adjacent pair of grades. The collapse for $M_{y,x}$ acts on the blocks between $|x|$ and $|y|$; on the remaining blocks it is the identity. This is the meaning of suspension in the following formulas. We define a pointed topologically enriched functor $K_{\mathcal{C}} \in \text{Fun}_+(\mathcal{J}, \mathbf{Top}_+)$ as follows.

First, we set

$$K_{\mathcal{C}}(i) = \bigvee_{|x|=i} S^{Nb}.$$

To define the morphisms, we use the $(\wedge, \text{Hom})$ adjunction. It suffices to provide maps:

$$g_{y,x} : \text{Hom}_{\mathcal{J}}(j, i) \wedge S^{Nb} \rightarrow S^{Nb}$$

for each pair $y, x \in \mathcal{C}$ with indices j, i . For fixed y and target degree, choose the tubular images for the different x disjoint, after stabilization; their simultaneous collapse is then a map to the wedge of target spheres. Let y and x be as above; we define the maps

$$S^{N(|y|-|x|)} \wedge (\mathbb{R}_{0 \leq}^{J_{y,x}})_+ \rightarrow Th(e_{y,x}) \rightarrow S^{N(|y|-|x|)},$$

where the first map is the collapse map of $e_{y,x}$ and the second is (the compactification of) the projection of $\varphi_{y,x}$ (see Eq. (28)). By suspending this map, we obtain:

$$g_{y,x} : S^{Nb} \wedge (\mathbb{R}_{0 \leq}^{J_{y,x}})_+ \longrightarrow S^{N(b-|y|+|x|)} \wedge Th(\nu e_{y,x}) \longrightarrow S^{Nb}.$$

Functoriality of $K_{\mathcal{C}}$ amounts to the equation

$$\partial_y g_{z,x} = g_{y,x} \circ g_{z,y},$$

which holds by Eq. (32): on a broken face the two collapses act on consecutive, disjoint Euclidean blocks. Choose tubular neighborhoods compatible with these products. This is the naturality check in [11, Proposition 2.24], specialized to normally framed categories with discrete object spaces.

For a flow category $\mathcal{C}$, we define $\mathcal{C}^{[b,a]}$ to be the full subcategory generated by objects $x \in \mathcal{C}$ satisfying the index condition

$$b \geq |x| \geq a.$$

We denote by $K_{[b,a]}$ the topological chain complex of $\mathcal{C}^{[b,a]}$. Note that there is a natural transformation

$$\beta_{b+1} : K_{[b,a]} \rightarrow K_{[b+1,a]}$$

induced by suspending further to $N(b+1)$; if $i = b+1$, we take $\beta_{b+1}(i)$ to be the zero map. Similarly, we have maps

$$\alpha_{a-1} : K_{[b,a-1]} \rightarrow K_{[b,a]}.$$

Definition 4.5. We extend Construction 4.4 to unbounded flow categories by taking the limit and colimit:

$$K_{\mathcal{C}} = \lim_a \text{colim}_b K_{[b,a]}.$$

Do the following def works? do we get a functor from $\mathcal{J}$?, see also the last todo

Construction 4.6. Let $\mathcal{C}$ be an embedded framed flow category bounded by b and 0. Using [Construction 4.4](#) and the based suspension spectrum functor Σ^∞ , we construct a functor $K'_\mathcal{C} \in \text{Fun}_+(\mathcal{J}, \mathbf{Sp}^O)$ with

$$K'_\mathcal{C}(i) = \bigoplus_{|x|=i} \mathbb{S}^{Nb}.$$

In $\mathbf{Sp}$, define the chain complex of $\mathcal{C}$ to be

$$K_\mathcal{C} := \Sigma^{-Nb} K'_\mathcal{C}$$

which has one copy of $\mathbb{S}$ for each object in the corresponding degree. A strict orthogonal-spectrum representative is obtained instead by applying $F_{\mathbb{R}^{Nb}}$ to the based-space diagram; this avoids treating desuspension as a literal inverse to suspension in the point-set category.

Proposition 4.7. *There is an equivalence of pointed ∞ -categories $N(\text{Ch}) \simeq N_{\mathbf{Top}}(\mathcal{J})$, where Ch is the indexing category of chain complexes (see [Definition 2.2](#)) and $N, N_{\mathbf{Top}}$ are the nerve and the topological nerve, respectively.*

Proof. The augmentation $\mathcal{J} \rightarrow \text{Ch}$ is the identity on integer objects, sends ∞ to the zero object, and collapses each positive-dimensional compactified orthant to the zero morphism. Composition is preserved because every composite through a strictly intermediate integer lies on such an orthant; identities and zero morphisms are preserved as well. This functor is a weak homotopy equivalence on every mapping space: for adjacent or equal integer objects it is the identity on S^0 , and the other mapping spaces are contractible. It is also essentially surjective. Apply the coherent nerve to the singular simplicial categories, whose mapping spaces are Kan complexes. The resulting equivalence is the enriched-category/nerve comparison of [\[20, Theorem 2.2.5.1\]](#). The same argument applies to each bounded restriction, retaining the zero object. $\square$

Proposition 4.8. *For $a \leq b$, localization of strict based enriched diagrams at objectwise stable equivalences gives*

$$\text{Fun}_+^{\text{str}}(\mathcal{J}_{[b,a]}, \mathbf{Sp}^O)[W_{\text{obj}}^{-1}] \simeq \text{Ch}_{[b,a]}(\mathbf{Sp}).$$

Thus the collapse diagram of a bounded framed flow category determines a coherent chain complex. The model structure and the interpretation of mapping spaces are specified in [Proposition 5.1](#).

By identifying 1-categories with their nerves, we end this section by motivating the next. Let $\mathcal{C}$ be a bounded, embedded, framed flow category and let $K \in \text{Fun}(\text{Ch}, \mathbf{Sp})$ be its coherent chain complex. By [Section 2](#), we obtain a spectral sequence $(E_r^{n,p}, d_r)$ where

$$E_1^{n,p} = \pi_{n-p} K_p = \bigoplus_{|x|=p} \pi_{n-p} \mathbb{S}.$$

Identifying stable homotopy groups with framed bordism via the isomorphism

$$\pi_{n-p} \mathbb{S} \simeq \Omega_{n-p}^{\text{fr}},$$

we observe that for $y, x \in \mathcal{C}$ with $|y| = p$ and $|x| = p - 1$, the first differential

$$d_1 : \Omega_{n-p}^{\text{fr}} \langle y \rangle \rightarrow \Omega_{n-p}^{\text{fr}} \langle x \rangle$$

is given by the formula

$$d_1(N) = N \times M_{y,x},$$

where $M_{y,x}$ is the morphism manifold in $\mathcal{C}$ and N is a representative for its bordism class. Survival to the second page corresponds to the existence of a flow category with objects s, y, x and morphism manifolds

$$M_{y,x}, \quad N = M_{s,y}, \quad \text{and} \quad W = M_{s,x}$$

such that their boundaries satisfy the relation

$$\partial W = M_{y,x} \times N.$$

5 Composition through a Bounded Chain Complex

****Note for rewriting.**** The endpoint/coend proof is retained in the body. Consider moving only the finite-dimensional representative lemma and its relative proof to an appendix if that improves the final organization; retain its precise statement and its use in the geometric realization argument here.

We model only composites

$$X_b \xrightarrow{f} K \xrightarrow{g} Y_a, \quad a \leq b,$$

where K is supported in $[b, a]$ and X_b, Y_a are concentrated in the indicated degrees. The endpoint spectra may be suspended spheres. This is the case needed for a right flow module followed by a left flow module. Intrinsically, the calculation is composition paired with a derived coend. The purpose of the model below is to compute this pairing by the ordinary union of the faces of an orthant. The cellular calculation in [Lemma 5.2](#) is what permits the ordinary coend; no cofibrancy assertion about arbitrary flow diagrams is needed.

5.1 Endpoint models

Write $\mathcal{A} = \mathcal{J}_{[b,a]}$, using the category of [Section 4](#). We work with based enriched functors and omit the zero object from the notation. Use compactly generated based spaces and the ordinary stable model structure on orthogonal spectra, not its positive variant; see [\[21, Theorem 9.2\]](#). A based space acts on a spectrum by tensoring; we write this action as $\wedge$. Cotensor by a based space T is denoted by Y^T .

Proposition 5.1. *The category $\mathcal{D} = \text{Fun}_+^{\text{str}}(\mathcal{A}, \mathbf{Sp}^O)$ has the projective stable model structure, with objectwise weak equivalences and fibrations. Its ∞ -categorical localization is*

$$\mathcal{D}_\infty = \mathcal{D}[W_{\text{obj}}^{-1}] \simeq \text{Ch}_{[b,a]}(\mathbf{Sp}).$$

The enriched mapping space from a projectively cofibrant diagram to an objectwise fibrant diagram computes the derived mapping space.

Proof. Apply [29, Theorem 7.2(1)] to modules over the opposite of the spectral category $\Sigma^\infty \mathcal{A}$. Its mapping spectra are cofibrant: the mapping spaces of $\mathcal{A}$ are based CW spaces, and suspension spectra are cofibrant in the chosen model structure. The projective structure is topological, so enriched maps between cofibrant and fibrant diagrams compute derived mapping spaces.

For the localization statement, use the monoidal comparisons of [29, Sections 6–7] to pass to symmetric spectra in simplicial sets. This is a combinatorial simplicial model category. The singular simplicial category $\text{Sing } \mathcal{A}$ has Kan mapping spaces, so the counit from the rigidification of its coherent nerve is a Dwyer–Kan equivalence. Thus [20, Proposition 4.2.4.4] identifies the localization of enriched diagrams with the functor ∞ -category out of that nerve. Now apply Proposition 4.7.

To pass to pointed diagrams, temporarily retain the zero object $*$. For an unrestricted diagram D , the zero morphisms give a natural map $\underline{D(*)} \rightarrow D$. On a projectively cofibrant diagram its inclusion is a cofibration: on a free cell this is the inclusion of the zero morphism in each mapping space, and the assertion follows by cell induction. Its quotient is strictly zero at $*$ and sends every zero morphism to zero. This quotient computes the objectwise homotopy cofiber. Hence reduction is an objectwise equivalence whenever $D(*)$ is contractible. Conversely, on strictly reduced diagrams the derived reduction is the identity. Reduction and inclusion therefore identify the localization of based diagrams with the full subcategory of functors taking $*$ to zero. $\square$

Here Map^h denotes a derived mapping *space*, and $\text{Nat}_{\mathcal{A}}$ denotes the space of strict enriched natural transformations. Choose a cofibrant representative of X , a fibrant representative of Y , and an objectwise cofibrant and fibrant representative of K . These choices are always available: for K , first take a projective cofibrant replacement and then a projective trivial cofibration to a fibrant diagram. Since the mapping spaces of $\mathcal{A}$ are cofibrant, projective cofibrations are objectwise cofibrations, so this last step preserves objectwise cofibrancy.

These replacements change representatives, not the morphisms in the localized category. We only use their universal comparison maps; no contractibility claim about the set of point-set replacements is required.

Define the covariant and contravariant weights

$$P(i) = \mathcal{J}(b+1, i), \quad B(i) = \mathcal{J}(i, a-1),$$

and the diagrams

$$Q_X(i) = P(i) \wedge X, \quad R_Y(i) = Y^{B(i)}.$$

There are objectwise stable equivalences

$$Q_X \longrightarrow X_b, \quad Y_a \longrightarrow R_Y. \tag{34}$$

At b and a , respectively, these maps are identities; at other objects the relevant orthant compactifications are based contractible.

Lemma 5.2. *If X is cofibrant, Q_X is projectively cofibrant. The weight B is built from finitely many free $\mathcal{A}^{\text{op}}$ -cells. Consequently,*

$$T(K) = B \otimes_{\mathcal{A}} K = \int^{i \in \mathcal{A}} B(i) \wedge K(i)$$

is cofibrant for objectwise cofibrant K , and T preserves objectwise stable equivalences between such diagrams.

Proof. Put $H_m = (\mathbb{R}_{0\leq}^m)_+$ and $D_m = (\partial\mathbb{R}_{0\leq}^m)_+$, with $H_0 = S^0$ and $D_0 = *$. For $m > 0$ this is a based disk–boundary pair, with the compactification point lying on the boundary. If $F_i(E)(j) = \mathcal{A}(i, j) \wedge E$, construct Q_X in decreasing order $i = b, \dots, a$ by attaching

$$F_i(D_{b-i} \wedge X) \longrightarrow F_i(H_{b-i} \wedge X).$$

The attaching subspace at i is the union of the faces supplied by higher degrees. At a lower degree $k < i$, the new piece is the face $t_i = 0$; its intersection with earlier pieces is exactly the indicated attaching subspace. Thus these attachments give Q_X .

Dually, build B in increasing order $i = a, \dots, b$ by attaching

$$\mathcal{A}(-, i) \wedge D_{i-a} \longrightarrow \mathcal{A}(-, i) \wedge H_{i-a}.$$

Tensoring over $\mathcal{A}$ with K turns this into $D_{i-a} \wedge K(i) \rightarrow H_{i-a} \wedge K(i)$ by enriched Yoneda. For objectwise cofibrant K these are cofibrations between cofibrant spectra. The resulting finite pushouts are homotopy pushouts, proving both assertions about T . $\square$

Proposition 5.3 (One-term sources and targets). *With the replacements above, there are equivalences*

$$\mathrm{Nat}_{\mathcal{A}}(Q_X, K) \simeq \mathrm{Map}_{\mathrm{Ch}_{[b,a]}}^h(\mathbf{Sp})(X_b, K), \quad \mathrm{Nat}_{\mathcal{A}}(K, R_Y) \simeq \mathrm{Map}_{\mathrm{Ch}_{[b,a]}}^h(\mathbf{Sp})(K, Y_a).$$

The strict maps on the left are precisely the enriched extensions of K to $\mathcal{J}_{[b+1,a]}$ with value X at $b+1$. Those on the right are the extensions to $\mathcal{J}_{[b,a-1]}$ with value Y at $a-1$. Explicitly, their extra structure maps are

$$f_i : P(i) \wedge X \longrightarrow K(i), \quad g_i : B(i) \wedge K(i) \longrightarrow Y,$$

compatible with the $\mathcal{A}$ -actions.

Proof. The source assertion follows from Lemma 5.2 and Eq. (34) and objectwise fibrancy of K . For the target assertion, take a projective cofibrant replacement $qK \rightarrow K$. Both diagrams are objectwise cofibrant, and the previous lemma gives a stable equivalence $T(qK) \rightarrow T(K)$ between cofibrant spectra. Since Y is fibrant, adjunction gives

$$\mathrm{Nat}_{\mathcal{A}}(K, R_Y) \cong \mathrm{Map}_{\mathbf{Sp}^o}(T(K), Y) \simeq \mathrm{Map}_{\mathbf{Sp}^o}(T(qK), Y) \cong \mathrm{Nat}_{\mathcal{A}}(qK, R_Y).$$

The last space computes the derived mapping space into R_Y , which is objectwise fibrant and equivalent to Y_a . The descriptions as endpoint extensions follow by writing out the enriched functor axioms. $\square$

5.2 The composition coend

Lemma 5.4. *Composition in $\mathcal{J}$ induces a homeomorphism*

$$B \otimes_{\mathcal{A}} P = \int^{i \in \mathcal{A}} \mathcal{J}(i, a-1) \wedge \mathcal{J}(b+1, i) \cong (\partial\mathbb{R}_{0\leq}^{\{a,\dots,b\}})_+.$$

This based space is homeomorphic to S^{b-a} .

Proof. The i -summand is the compactified face $t_i = 0$. For $j > i$, the coend relation on

$$B(i) \wedge \mathcal{A}(j, i) \wedge P(j)$$

identifies the two copies of the intersection $t_i = t_j = 0$. An intersection of more than two faces is indexed by their union of labels; the pairwise identifications agree there by associativity. For $a = b$ there is just one summand S^0 and no nontrivial relation. The induced map is a continuous bijection from a compact quotient to the Hausdorff boundary compactification, hence a homeomorphism.

For $m = b - a + 1$, the map

$$t \mapsto t - \frac{1}{m} \left(\sum_i t_i \right) (1, \dots, 1)$$

identifies $\partial \mathbb{R}_{\leq}^m$ with the hyperplane $\sum_i u_i = 0$; its inverse subtracts the minimum coordinate. Fix an ordered linear identification of this hyperplane with $\mathbb{R}^{b-a}$ when identifying its compactification with a sphere. $\square$

Proposition 5.5 (Composition through a bounded complex). *Let $a \leq b$ and let*

$$X_b \xrightarrow{f} K \xrightarrow{g} Y_a$$

be morphisms in $\text{Ch}_{[b,a]}(\mathbf{Sp})$. Use the replacements and representatives f_i, g_i of [Proposition 5.3](#). Under the equivalence

$$\text{Map}_{\text{Ch}_{[b,a]}(\mathbf{Sp})}^h(X_b, Y_a) \simeq \text{Map}_{\mathbf{Sp}}^h(\Sigma^{b-a} X, Y),$$

the composite is represented by

$$(B \otimes_{\mathcal{A}} P) \wedge X \longrightarrow Y \tag{35}$$

whose restriction to the i -face is

$$B(i) \wedge P(i) \wedge X \xrightarrow{1 \wedge f_i} B(i) \wedge K(i) \xrightarrow{g_i} Y. \tag{36}$$

Proof. The diagram Q_X is projectively cofibrant and R_Y is objectwise fibrant. Their enriched mapping space therefore computes the derived maps from X_b to Y_a . Coend–cotensor adjunction identifies it with

$$\text{Map}_{\mathbf{Sp}^o}((B \otimes_{\mathcal{A}} P) \wedge X, Y).$$

Apply [Lemma 5.4](#). The composite is represented in the localization by

$$X_b \xleftarrow{\sim} Q_X \xrightarrow{\tilde{g}\tilde{f}} R_Y \xleftarrow{\sim} Y_a.$$

The adjoint of its middle map is [Eq. \(35\)](#), and naturality is precisely agreement on the face intersections. $\square$

Lemma 5.6 (Compatibility with unreplaced representatives). *Let X be cofibrant, let K_0 be objectwise cofibrant, and suppose strict endpoint data give maps*

$$Q_X \longrightarrow K_0 \longrightarrow R_{Y_0}.$$

Their ordinary coend composite represents the composite of the corresponding localized maps, even when K_0 and Y_0 are not fibrant.

Proof. Choose a fibrant replacement $Y_0 \rightarrow Y$ and a projective trivial cofibration $K_0 \rightarrow K$ with K fibrant. The induced map $K_0 \rightarrow R_Y$ extends to $K \rightarrow R_Y$ by lifting, since R_Y is projectively fibrant. Composing $Q_X \rightarrow K_0 \rightarrow K$ with this extension gives exactly the original coend map followed by $Y_0 \rightarrow Y$. Apply [Proposition 5.5](#). Objectwise cofibrancy of K is preserved in this replacement. $\square$

In the geometric application, retain the finite-dimensional collapse maps of [Construction 4.4](#) before making replacements. Applying the free orthogonal spectrum functor $F_{\mathbb{R}^N}$ to a common suspended based-space diagram gives objectwise cofibrant representatives: $F_{\mathbb{R}^N} S^{N+t}$ represents $\Sigma^t \mathbb{S}$ stably. [Lemma 5.6](#) then applies. The free functors and their stable comparison with suspension spectra are part of the orthogonal-spectrum model of [\[21, Section 2\]](#).

Lemma 5.7 (Finite-dimensional endpoint representatives). *Suppose K_0 comes from a finite based-space $\mathcal{A}$ -diagram of finite wedges of spheres, by applying $F_{\mathbb{R}^N}$. An endpoint morphism from a suspended sphere to K_0 admits, after further suspension, simultaneous based-space representatives of its maps f_i . One may prescribe representatives on an initial segment of the cellular construction of Q_X , provided their stable class is the restriction of the given morphism. Homotopies admit the same relative construction.*

Proof. Write $X = \Sigma^d \mathbb{S}$ and $K_0 = F_{\mathbb{R}^N} Z$. At suspension level q , use the source $P \wedge S^{N+q+d}$ and target diagram $\Sigma^q Z$. Mapping out of the finite free-cell filtration of P expresses the space of endpoint maps as a finite iterated homotopy pullback. Indeed, at each stage the restriction map is a fibration of mapping spaces, since the attaching pair is a cofibration; enriched Yoneda identifies the other terms with ordinary based mapping spaces into $\Sigma^q Z(i)$.

After stabilization, these terms compute the mapping spaces into $\Sigma^{-N} \Sigma^\infty Z(i)$, which represents $K_0(i)$. Filtered colimits of spaces commute with finite homotopy limits [\[20, Proposition 5.3.3.3\]](#). Consequently the same finite pullback computes the derived endpoint mapping space after taking the colimit in q . Every component, and every homotopy between components, is therefore represented at a finite level.

For prescribed data on an initial segment, take the homotopy fiber of restriction over those data. A chosen stable extension, together with a homotopy identifying its restriction with the prescribed map, gives a point of this fiber. The same finite-limit argument supplies it at a finite suspension level. The homotopy extension property then makes the restriction exact. This proves the relative assertion as well as its version for homotopies. It does not assert that the space of extensions is contractible. $\square$

For the differential, the specialization is

$$b = p, \quad a = p - r, \quad X = \Sigma^{n-p} \mathbb{S}, \quad Y = K_{p-r-1}.$$

The lower endpoint data come from $K_{[p, p-r-1]}$. The coend produces

$$\Sigma^{n-p+r} \mathbb{S} \longrightarrow K_{p-r-1},$$

hence an element of $E_1^{n-1, p-r-1}$. Identifying its class on E_{r+1} with the differential requires the connecting-map comparison in [Proposition 2.32](#); the outstanding G2 check is recorded after [Proposition 6.4](#). The next section proves the geometric comparison with this coend map.

6 Differential in a flow category

We describe the differential by composing a right flow module with the fixed flow manifolds. All flow manifolds, module manifolds, and bordisms below are compact; all boundary identifications include their higher intersections and compatible normal framings.

The geometric input is relative Pontryagin–Thom on finite diagrams of faces. We give the rounding and collapse comparison first, then realize algebraic extensions by relative transversality. Only framed bordism classes are needed; we do not assert a classification of flow bimodules or contractibility of the space of framings.

Proof verification needed. Status of the differential theorem. The geometric realization, rounding, and corner-restoration arguments below have been checked and written out. The identification with the spectral-sequence differential still depends on the G2 comparison recorded after [Proposition 6.4](#). The concluding argument records that dependency; it is not a claim that G2 has been resolved.

Theorem 6.1. *Let $\mathcal{C}$ be a flow category with finitely many objects in each degree and compact morphism manifolds, equipped with compatible neat embeddings and stable normal framings. Let $K_{\mathcal{C}}$ be its associated coherent chain complex and let $(E_r^{n,p}, d_r)$ be the spectral sequence of $\mathcal{I}K_{\mathcal{C}}$. Then*

$$E_1^{n,p} = \bigoplus_{|x|=p} \Omega_{n-p}^{\text{fr}}.$$

Fix integers $n \geq p$ and $r \geq 0$, and let $L = (L_x)_{|x|=p}$ be closed framed $(n-p)$ -manifolds representing $\ell \in E_1^{n,p}$.

1. The element ℓ survives to E_{r+1} , in the sense of [Definition 2.17](#), if and only if L extends to a degree- n neatly embedded, normally framed right flow module W over $\mathcal{C}_{[p,p-r]}$.
2. Given such W , for each object z of degree $p-r-1$ form the closed framed $(n-p+r)$ -manifold

$$T_z(W) = M_{-,z} \circ W = \text{round} \left(\bigcup_{\substack{x \in \mathcal{C} \\ p \geq |x| \geq p-r}} W_x \times M_{x,z} \right).$$

The union denotes gluing along the product faces of [Definition 6.2](#). The collection

$$\tau(W) = ([T_z(W)])_{|z|=p-r-1} \in E_1^{n-1,p-r-1}$$

lies in $\mathcal{E}_{r+1}^{n-1,p-r-1}$ and represents

$$d_{r+1}^{n,p}([\ell]_{r+1}) = [\tau(W)]_{r+1}.$$

This page class is independent of W ; the class $\tau(W)$ in E_1 need not be.

3. The differential $d_{r+1}^{n,p}([\ell]_{r+1})$ vanishes if and only if there is a choice of W extending L for which every $T_z(W)$ is framed null-bordant. For that choice, null-bordisms with the prescribed corner faces extend W over $\mathcal{C}_{[p,p-r-1]}$.

The constructions use only bounded grading windows, which are finite under the stated hypothesis. No new unbounded realization construction is asserted here: $K_{\mathcal{C}}$ is the given coherent chain complex, and the argument applies to its finite restrictions.

6.1 Gluing a right and a left module

Definition 6.2. Let $a \leq b \leq n$, let $\mathcal{C}$ be a finite framed flow category supported in $[b, a]$, and let W, V be compact framed right and left flow modules over $\mathcal{C}$ of degrees n and a , respectively. Set $I = \{a, \dots, b\}$. For a nonempty subset $S = \{i_0 > \dots > i_k\} \subseteq I$, define

$$D_S(W, V) = \bigsqcup_{|x_j|=i_j} W_{x_0} \times M_{x_0, x_1} \times \dots \times M_{x_{k-1}, x_k} \times V_{x_k}.$$

For $k = 0$ this means $\bigsqcup_{|x|=i_0} W_x \times V_x$. Flow composition and the module actions give a diagram indexed by nonempty subsets with arrows $T \rightarrow S$ for $S \subseteq T$. Write

$$|D(W, V)| = \operatorname{colim}_{\emptyset \neq S \subseteq I} D_S(W, V).$$

We denote its framed rounding, constructed in [Lemma 6.3](#), by $V \circ W$.

Lemma 6.3 (The manifold and its framing). *For the modules of [Definition 6.2](#), $|D(W, V)|$ admits a rounding to a closed smooth manifold of dimension $n - a$. The compatible normal framings determine a stable framing of this rounding. Its framed bordism class is independent of the compatible collars and rounding choices.*

Proof. Put $d = n - a$. The stratum D_S has dimension $d + 1 - |S|$. Boundary exhaustion and associativity give, near its interior, the model

$$\mathbb{R}^{d+1-|S|} \times \partial \mathbb{R}_{0 \leq}^{|S|}.$$

Indeed, the $|S|$ top pieces are its incident coordinate faces, and their intersections are exactly the products indexed by supersets of S . Thus the colimit has no unpaired boundary.

Choose product collars, compatibly from the deepest strata outwards. Relative collars and framed composition are supplied by [\[26, Lemmas 3.19, 4.9, and 4.41–4.43\]](#) (arXiv version 3). In that construction the pieces $W_x \times V_x$ acquire an extra collar direction. On a double overlap the two collar coordinates are exchanged; their zero faces are precisely the identifications in $D(W, V)$. Associativity gives the same description on higher overlaps. A nearby regular hypersurface therefore rounds this particular colimit.

For our normal-framing convention, stabilize the compatible product embeddings into the coordinate faces of a common orthant. On an overlap both normal trivializations restrict to the same ordered product trivialization on D_S . The exchanged collar coordinates are tangent to the thickened ambient orthant, not additional normal-framing factors. Rounding these coordinates in both the ambient space and the submanifold preserves that normal trivialization. Restricting to the rounded ambient boundary gives the asserted framed manifold. For independence, join two compatible systems of collars by the relative isotopy of [\[26, Lemmas 3.19 and 4.9\]](#). Extend the stabilized embeddings over this one-parameter family, relative to its ends, using [\[14, Proposition 2.8\]](#). Product tubular neighborhoods extend after shrinking their radii. Rounding the resulting family gives a framed bordism between the two choices, as in [\[26, Lemmas 4.42–4.43\]](#). Only auxiliary choices vary; the framing data are fixed. $\square$

Proposition 6.4 (Collapse and composition). *Let $\mathcal{C}, W, V$ be as in [Definition 6.2](#), with compatible neat embeddings and normal framings. Their collapse data determine morphisms*

$$f_W : (\Sigma^{n-b}\mathbb{S})_b \longrightarrow K_{\mathcal{C}}, \quad g_V : K_{\mathcal{C}} \longrightarrow \mathbb{S}_a$$

in $\mathrm{Ch}_{[b,a]}(\mathbf{Sp})$. Under [Proposition 5.5](#), their composite corresponds to

$$\mathrm{PT}(V \circ W) : \Sigma^{n-a}\mathbb{S} \longrightarrow \mathbb{S}.$$

Proof. Choose product collars and tubular neighborhoods by induction over faces, retaining the choices already made on their intersections. Extend a product metric from a smaller boundary collar and use a normal exponential neighborhood; compactness permits a common smaller radius. Relative neat embeddings exist after stabilization [\[14, Proposition 2.8\]](#); compare [\[11, Appendix A\]](#). There are finitely many pieces, so these choices give compatible finite-dimensional collapse maps. The module boundary identities are the enriched endpoint identities of [Proposition 5.3](#). Keep these maps in their suspended form, apply a free orthogonal spectrum functor, and use [Lemma 5.6](#) to pass to the fibrant representatives used for derived mapping spaces.

Here is the local comparison before replacement. For nonempty $S \subseteq I$, embed D_S in

$$E_S = \mathbb{R}^N \times \mathbb{R}_{0 \leq}^{I \setminus S}.$$

Its normal bundle ν_S has rank $c = N + |I| - (n - a) - 1$, independent of S . Choose tubular neighborhoods which are products on all collars. Restriction to a face then identifies both the normal bundles and their collapse maps, giving a diagram

$$E_S^+ \longrightarrow \mathrm{Th}(\nu_S).$$

Order the face diagram from deepest intersections to the top faces. Each latching inclusion is an inclusion of a union of boundary faces, hence a cofibration by the product collars. The same holds for the disk and sphere bundles of the normal bundles. Thus the ordinary face colimits of the compactified ambient spaces and Thom spaces compute their homotopy colimits. The ambient colimit is $\mathbb{R}^N \times \partial \mathbb{R}_{0 \leq}^I$, whose compactification is $S^{N+|I|-1}$. On each collar the embedding and tubular neighborhood are products, so the transition maps of the normal bundles restrict to the same maps on every intersection. They assemble to a bundle ν over $|D(W, V)|$, and

$$\mathrm{colim}_S \Sigma^\infty \mathrm{Th}(\nu_S) \simeq \mathrm{Th}_{\mathbf{Sp}}(\nu).$$

This target identification can also be obtained from preservation of colimits by the Thom-spectrum functor [\[3, Section 3.3\]](#), after regarding the bundle data as a diagram of stable spherical fibrations.

In a product tubular chart, rounding changes only the base collar coordinates and leaves the normal coordinate unchanged. Outside these charts every collapse sends the complement of the tubular neighborhood to the basepoint. Consequently these local identifications give a homotopy from the assembled collapse to the collapse of the rounded embedding, relative to the unchanged regions. After using the framing to map the Thom space to S^c , its restriction to the i -face is the product collapse for $\bigsqcup_{|x|=i} W_x \times V_x$. These are exactly the face maps in [Eq. \(36\)](#); the coend relations identify their common restrictions. Thus the assembled map is the categorical composite. $\square$

Proof verification needed. G2 — unresolved comparison, with related references. The proposition proves equality between the geometric collapse and the coend composite, using the same ordered boundary sphere. To identify this with the differential defined from the filtration, one must still compare that sphere identification with the connecting map used in [Proposition 2.32](#). Lurie [19, Construction 1.2.2.6 and Definition 1.2.2.9] supplies the filtered differential, and Côté–Kartal [11, Proposition 2.24] supplies collapse naturality; neither states this comparison for our endpoint model. The proof of [Proposition 2.32](#) also passes from a suspended lower chain to an index-shifted chain without identifying the resulting coherent map. Thus the outstanding task is to compare the maps, including the suspension sign, not merely to choose an orientation. The one-degree case gives the product framing, but does not settle all grading windows. No exact reference for this remaining identification has been located.

6.2 Realizing extensions and restoring corner faces

Here realization means representing an endpoint morphism by a right flow module whose Pontryagin–Thom collapse gives that morphism.

Lemma 6.5 (Realization with a fixed top representative). *Let $\mathcal{C}$ be a finite neatly embedded, normally framed flow category supported in $[p, a]$, let $n \geq p$, and let*

$$f : (\Sigma^{n-p}\mathbb{S})_p \longrightarrow K_{\mathcal{C}}$$

be a morphism of coherent chain complexes. Suppose its component in degree p is represented by the Pontryagin–Thom maps of closed framed manifolds $(L_x)_{|x|=p}$. After stabilization, f is represented by a compact neatly embedded, normally framed right flow module W of degree n with $W_x = L_x$ in degree p .

Proof. Put $d = n - p$. We construct the pieces in decreasing degree, maintaining a homotopy between their endpoint collapse data and the corresponding restriction of f . At degree p , the prescribed Pontryagin–Thom representatives L_x and the hypothesis provide this homotopy. Homotopy extension along the cells of $Q_{\Sigma^d\mathbb{S}}$ transports the remaining endpoint data when a partial representative is changed.

Suppose that the pieces above i have been constructed. For $|x| = i$, their products $W_y \times M_{y,x}$ prescribe the boundary of the next piece, with all intersections fixed. Use ordered Euclidean blocks as in [Construction 4.4](#). After stabilization, write their boundary collapse as a map

$$(\partial\mathbb{R}_{0\leq}^{p-i})_+ \wedge S^{c_i+d} \longrightarrow \bigvee_{|x|=i} S^{c_i},$$

where c_i is the normal rank in the active coordinates. For a face indexed by y , the normal coordinates are the ordered sum of those for W_y and $M_{y,x}$. In particular, the inverse image of a chosen non-basepoint in the x -sphere is the actual product $W_y \times M_{y,x}$, with its product framing.

The endpoint morphism f , transported along the partial homotopy, supplies a stable extension of this boundary map over $(\mathbb{R}_{0\leq}^{p-i})_+ \wedge S^{c_i+d}$, including its relative homotopy class. The relative finite-cell argument of [Lemma 5.7](#) realizes this extension after a finite suspension. Here any additional suspension is placed in extra module coordinates, leaving the flow coordinates fixed. Thus it suspends all previously constructed module collapses by an identity and preserves the

prescribed product boundary. Homotopy extension makes the boundary restriction exact, and the compatible collars make the extension a product near every face.

Smooth the map near the inverse images of the chosen non-basepoints and make it transverse, keeping a smaller boundary collar fixed. The perturbation is supported away from that collar; there the domain has no boundary, so ordinary relative transversality applies. Compactness of the inverse images follows from the based map on the compactified domain. The inverse image defining W_x has dimension

$$(c_i + d) + (p - i) - c_i = n - i.$$

The derivative gives its normal framing, agreeing on the collar with the prescribed framing. The relative Pontryagin–Thom homotopy identifies its collapse with the chosen extension while fixing that collar. This is the inverse-image argument of [14, Theorem 3.17 and Section 9], with the perturbation and homotopy performed relative to a region already in product form. Extending this homotopy over the remaining cells preserves the class of the entire morphism f .

The construction uses only finitely many pieces and suspensions. Increase the Euclidean blocks to a common size and extend embeddings relative to the boundary using [14, Proposition 2.8]. The unused face coordinates above p are placed at positive constants; their extra normal directions carry the trivial framing. This gives the embeddings of Definition 3.26, retaining the original L_x in degree p . $\square$

Lemma 6.6 (Restoring the faces of a null-bordism). *Let W be a degree- n framed right flow module over $\mathcal{C}_{[p,a]}$ and let z have degree $a - 1$. Suppose the framed rounding $T_z(W)$ is null-bordant. Then a framed null-bordism can be given corners and, after stabilization, a neat embedding extending the prescribed boundary data, so that it defines a manifold W_z of dimension $n - a + 1$ with faces*

$$\partial_i W_z = \bigsqcup_{|x|=i} W_x \times M_{x,z}, \quad a \leq i \leq p,$$

and with the higher intersections prescribed by flow composition and the module action. Performing this construction for all z of degree $a - 1$ extends W over $\mathcal{C}_{[p,a-1]}$.

Proof. Let B_z be a framed null-bordism together with its framed boundary identification $\partial B_z \cong T_z(W)$. Retain the collar region between the original cornered union and its rounding in Lemma 6.3. Attach its rounded end to a boundary collar of B_z . The other end has precisely the prescribed faces and their original intersections; no choices of new face identifications are involved.

Use the outward-normal-first boundary convention. The two collars being attached have opposite outward normals, so the attaching map reverses that line and identifies the remaining framed directions by the given framed boundary identification. Extend over a product collar. This transports the stable framing of B_z to the manifold with restored corners and agrees with every prescribed product framing. Choose a neat embedding relative to this entire boundary collar after stabilization [14, Proposition 2.8]. Its stable normal framing is the complement of the transported stable tangent framing; a further stabilization realizes the prescribed boundary homotopy of normal trivializations on a collar. Distinct objects in degree $a - 1$ impose no additional identities on one another. $\square$

6.3 Proof of the differential theorem

Proof of Theorem 6.1, subject to the G2 comparison. The degree- p spectrum is a wedge of sphere spectra, giving the stated first page by Pontryagin–Thom.

If L extends to W over $\mathcal{C}_{[p,p-r]}$, its collapse data give

$$f_W : (\Sigma^{n-p}\mathbb{S})_p \longrightarrow K_{\mathcal{C}_{[p,p-r]}}$$

extending the class of L . Proposition 2.24 implies survival to E_{r+1} . Conversely, that proposition supplies an algebraic extension for every surviving class; Lemma 6.5 realizes it with the prescribed top manifolds L .

Put $a = p - r$. For each z of degree $a - 1$, the manifolds $M_{x,z}$ form a left module of degree a over $\mathcal{C}_{[p,a]}$. Its collapse data are the lower endpoint data of $K_{\mathcal{C}}$ at z . By Proposition 2.32, composing with f_W gives the z -component of the raw obstruction $\partial_{r+1}(f_W)$. Proposition 6.4 identifies this component with $[T_z(W)]$. The same algebraic differential description shows that this collection belongs to $\mathcal{E}_{r+1}$ and represents $d_{r+1}([\ell]_{r+1})$.

Finally suppose this page differential is zero. Lemma 2.33, applied with $r + 1$, supplies an extension f' of the same top class for which

$$\partial_{r+1}(f') = 0 \quad \text{in } E_1^{n-1,p-r-1}.$$

Realize f' by a module W' using Lemma 6.5. Its Toda manifolds are framed null-bordant by the composition comparison. Apply Lemma 6.6 to extend W' one degree further. Conversely, null-bordant Toda manifolds give zero raw obstruction and hence zero page differential. The replacement of W by W' is essential: vanishing on the page does not assert null-bordism for every initial choice of W . $\square$

To add to the paper. Add a direct geometric description of $E_r^{n,p}$: it is the group of level- p framed collections W_p that extend to degree- n right flow modules over $\mathcal{C}_{[p,p-r+1]}$, modulo the level- p classes obtained by gluing degree- $(n + 1)$ right flow modules V over the upper window $\mathcal{C}_{[p+r-1,p+1]}$ to the fixed flow manifolds. At an object x of index p , this glued collection is $\bigcup_{p+1 \leq |y| \leq p+r-1} V_y \times M(y, x)$; these classes encode the cumulative incoming-boundary relations through d_{r-1} .

7 Tangential structures and change of rings

In this section we point out how to generalize Theorem 6.1 to a case where we replace the framing structure on the flow category $\mathcal{C}$ by other tangential structures. As a result we get a spectral sequence that calculates the bordism homology $\Omega_*^{\mathcal{X}}(|\mathcal{C}|)$.

To start with, we will generalize the definition of normal framing.

Definition 7.1. A tangential structure is a collection of fibrations $\mathcal{X}(n) \rightarrow BO(n)$, with maps $\mathcal{X}(n) \rightarrow \mathcal{X}(n + 1)$ such that

$$\begin{array}{ccc} \mathcal{X}(n) & \longrightarrow & \mathcal{X}(n + 1) \\ \downarrow & & \downarrow \\ BO(n) & \longrightarrow & BO(n + 1) \end{array}$$

Definition 7.2. Let $M \hookrightarrow \mathbb{R}_{0\leq}^k \times \mathbb{R}^N$ be an embedded m dimensional $\langle k \rangle$ manifold. A **normal $\mathcal{X}$ structure on M** is a choice of a lift

$$\begin{array}{ccc} & & \mathcal{X}(N) \\ & \nearrow \nu \tilde{M} & \downarrow \\ M & \xrightarrow{\nu M} & \underline{BO(N + k - m)} \end{array} \quad (37)$$

where the diagram takes place in the category $\langle k \rangle$ spaces, which is a functorial $[\tilde{1}]^k$ family of lifts

$$\begin{array}{ccc} & & \mathcal{X}(N) \\ & \nearrow & \downarrow \\ M(A) & \xrightarrow{\nu M(A)} & \underline{BO(N + k - m)} \end{array} \quad (38)$$

for each $A \subseteq [\tilde{1}]^k$.

Definition 7.3. We say that an $\mathcal{X}$ structure is **multiplicative** if it is equipped with structure maps

$$\mathcal{X}(n) \times \mathcal{X}(m) \rightarrow \mathcal{X}(n + m)$$

that commutes with the maps

$$BO(n) \times BO(m) \rightarrow BO(n + m)$$

Note that if $\mathcal{X}$ is multiplicative then a choice of $\mathcal{X}$ structure on two embedded manifolds M, L induces a $\mathcal{X}$ structure on $M \times L$ in the following way

$$\begin{array}{ccccc} & & \mathcal{X}(d) \times \mathcal{X}(d') & \longrightarrow & \mathcal{X}(d + d') \\ & \nearrow & \downarrow & & \downarrow \\ L \times M & \longrightarrow & \underline{BO(d) \times BO(d')} & \longrightarrow & \underline{BO(d + d')} \end{array}$$

where d and d' are the codimensions of L and M , respectively.

For an tangential structure $\mathcal{X}$ we denote by ζ_n the universal real vector bundle over $X(n)$, which is (by abuse of notation) the pullback of the universal real vector bundle $\zeta_n \rightarrow BO(n)$. Let M be an $\mathcal{X}$ structure manifold (for simplicity - a closed manifold) and denote by d the codimension of M in the ambient space, we obtain the following diagram:

$$\begin{array}{ccccc} \nu M & \longrightarrow & \zeta_n & \longrightarrow & \zeta_n \\ \downarrow & \lrcorner & \downarrow & \lrcorner & \downarrow \\ M & \xrightarrow{\nu \tilde{M}} & X(d) & \longrightarrow & BO(d) \end{array}$$

where the right square is pullback by definition and the same for the whole rectangle. We denote

$$\varphi_M: \nu M \rightarrow \zeta_n \quad (39)$$

Note that although we omitted $\nu \tilde{M}$ from the notation, a different choices of $\nu \tilde{M}$ defines different map φ_M . Note that if M is $\langle k \rangle$ manifold then φ_M is a map of $\langle k \rangle$ vector bundles.

Given a multiplicative structure we can define an $\mathcal{X}$ structured flow category (compare to [Definition 3.23](#)).

Definition 7.4. Let $\mathcal{C}$ be an embedded flow category, an $\mathcal{X}$ structure on $\mathcal{C}$ is a choice of $\mathcal{X}$ structure on each embedded morphism manifold $M_{y,x}$ such that

$$\partial_y \varphi_{z,x} = \varphi_{z,y} \oplus \varphi_{y,x}$$

where $\varphi_{y,x}$ is the isomorphism of $\langle J_{y,x} \rangle$ bundles arise from the $\mathcal{X}$ structure on $M_{y,x}$, see Eq. (39).

****Note for rewriting.**** Provide enough multiplicative structure on $M\mathcal{X}$ to define the module-valued realization below. An E_∞ -ring structure is a safe sufficient framework; investigate whether a weaker structure, in the sense suggested by Lurie’s term “multiplicative cohomology theory” (*Chromatic Homotopy Theory*, Lecture 1, [18]), suffices. The multiplicative structure enters at the definition stage: it supplies $\text{Mod}_{M\mathcal{X}}$ and the coherent composition used in the realization. Cohen assumes that E is represented by a commutative ring spectrum (explicitly meaning an E_∞ -ring) for the corresponding Floer E -theory construction; see [10, Sec. 1, p. 2]. Check that the current maps $\mathcal{X}(n) \times \mathcal{X}(m) \rightarrow \mathcal{X}(n+m)$ provide the required coherences.

Similarly to Construction 4.4, using Pontryagin Thom we get a coherent chain complex in $\text{Mod}_{M\mathcal{X}}$ where $M\mathcal{X}$ is the Thom spectrum of $\mathcal{X}$.

Construction 7.5. Let $\mathcal{C}$ be a bounded $\mathcal{X}$ flow category. For each $x, y \in \mathcal{C}$ we define a map

$$g_{y,x} : S^{N(|y|-|x|)} \wedge (\mathbb{R}_{0 \leq}^{J_{y,x}})_+ \rightarrow Th(\nu M_{y,x}) \rightarrow Th(\zeta_{N(|y|-|x|)}).$$

where the first map is the collapse map of the embedding $e_{y,x}$ (see Definition 3.4) and the second map is $Th(\varphi_{y,x})$. The coherence of being $\mathcal{X}$ flow categories amounts to

$$g_{z,y} \wedge g_{y,x} = \partial_y g_{z,x}$$

as maps between diagrams in **Top**.

By taking $\Sigma^{N(|y|-|x|)} \Sigma_+^\infty g_{y,x}$ we get a collection of maps to the Thom spectrum of $\mathcal{X}$

$$\mathbb{S} \otimes (\mathbb{R}_{0 \leq}^{J_{y,x}})_+ \rightarrow M\mathcal{X}.$$

which is equivalent to a map

$$(\mathbb{R}_{0 \leq}^{J_{y,x}})_+ \rightarrow \text{Hom}_{\mathbf{Sp}^O}(\mathbb{S}, M\mathcal{X}),$$

the last step is to use the adjunction between $\mathbb{S} - \text{Mod}$ and $M\mathcal{X} - \text{Mod}$ to obtain a map

$$\bar{g}_{y,x} : (\mathbb{R}_{0 \leq}^{J_{y,x}})_+ \rightarrow \text{Hom}_{M\mathcal{X}}(M\mathcal{X}, M\mathcal{X}).$$

To get that the collection of these maps define a functor

$$\mathcal{J} \rightarrow \text{Mod} - \mathcal{X}$$

we use the following commutative diagram:

$$\begin{array}{ccc} \text{Hom}_{\mathbf{Top}}(S^d, \zeta_d) \times \text{Hom}_{\mathbf{Top}}(S^{d'}, \zeta_{d'}) & \xrightarrow{\wedge} & \text{Hom}_{\mathbf{Top}}(S^{d+d'}, \zeta_{d+d'}) \\ \downarrow (\Sigma^{-d} \Sigma_+^\infty, \Sigma^{-d'} \Sigma_+^\infty) & & \downarrow \Sigma^{-(d+d')} \Sigma_+^\infty \\ \text{Hom}_{\mathbf{Sp}}(\mathbb{S}, M\mathcal{X}) \times \text{Hom}_{\mathbf{Sp}}(\mathbb{S}, M\mathcal{X}) & & \text{Hom}_{\mathbf{Sp}}(\mathbb{S}, M\mathcal{X}) \\ \downarrow \text{Mod} & & \downarrow \text{Mod} \\ \text{Hom}_{M\mathcal{X}}(M\mathcal{X}, M\mathcal{X}) \times \text{Hom}_{M\mathcal{X}}(M\mathcal{X}, M\mathcal{X}) & \xrightarrow{\circ} & \text{Hom}_{M\mathcal{X}}(M\mathcal{X}, M\mathcal{X}) \end{array}$$

I am confuse, we can move between strict functors of topologically enriched categories to ∞ functors in $\mathbf{Sp}$, but can we do the same for $M\mathcal{X}$ modules?

Here is a question: Let $E \in \mathbf{Sp}^O$ do we have a category of E modules and adjunction?

where Mod denote the $\mathbb{S} - \text{Mod}, \mathcal{X} - \text{Mod}$ adjunction. We obtain that

$$\bar{g}_{z,y} \otimes \bar{g}_{y,x} = \partial_y \bar{g}_{z,x}$$

hence we got a (topologically enriched) coherent chain complex $\text{Fun}(\mathcal{J}, \text{Mod} - \mathcal{X})$.

The definition of a right flow module (Definition 3.24) is modified by replacing framed flow modules by $\mathcal{X}$ structured flow modules and Ω^{fr} by $\Omega^{\mathcal{X}}$.

Similarly, we define when two $\mathcal{X}$ flow categories are composable. To define the composition category we modify Definition 6.2 we get a boundary diagram

$$D: [\tilde{1}]^{\{b, \dots, a\}} \setminus \underline{0} \rightarrow \mathbf{Top}$$

such that $D(\{a\})$ are $\mathcal{X}$ manifolds with $\langle A \setminus \{a\} \rangle$ corner structure. he lifts of the structure map $\nu D(B)$ assemble into a lift of the structure map $\nu \text{colim } D$ hence induces a $\mathcal{X}$ structure on the colimit. We get that

Corollary 7.6. *Let $\mathcal{C}, \mathcal{D}$ be two composable flow categories bounded by $b+1, a-1$, with $\mathcal{X}$ structure and flow manifolds written as $M_{-, -}, N_{-, -}$ then*

$$\bigcup_{b \leq p \leq a} M_{b+1, p} \times N_{p, a-1} := \text{colim } D$$

is an $\mathcal{X}$ manifold (compare to Definition 6.2). Where D is defined in the following way: for each $B = \{i_k, \dots, i_1\} \subseteq \{b, \dots, a\}$

$$D(i_k, \dots, i_1) = M_{b+1, i_k} M_{i_k, i_{k-1}} \times \dots M_{i_2, i_1} \times N_{i_1, a-1}.$$

We write

$$M_{b+1, a-1} := \bigcup_{b \leq p \leq a} M_{b+1, p} \times N_{p, a-1} := \text{colim } D.$$

We define composition category as same as the framed case.

Theorem A. *Let $\mathcal{C}$ be a flow category with an $\mathcal{X}$ -tangential structure. There is a spectral sequence:*

$$E_1^{n, p} = \bigoplus_{|z|=p} \Omega_{n-p}^{\mathcal{X}}$$

which conditionally converges to the homotopy groups of the realization:

$$E_r^{n, p} \implies \pi_n |\mathcal{C}|.$$

An element

$$W_p = \{W_z\}_{|z|=p} \in E_1^{n, p}$$

survives to $E_{r+1}^{n, p}$ if and only if the collection W_p extends to a right flow module over $\mathcal{C}_{[p, p-r]}$ (see Definition 3.28). The differential is represented geometrically by the Toda manifolds:

$$d_{r+1}(W_p) = \bigcup_{y \in \mathcal{C}_{p-(r+1)}} (\mathcal{M}_{-, y} \circ W) \in \bigoplus_{y \in \mathcal{C}_{p-(r+1)}} \Omega_{n-p+r}^{\mathcal{X}} = E_{r+1}^{n-1, p-(r+1)}$$

Remark 7.7. The convergence of the spectral sequence in Theorem A relies on the convergence results established for chain complexes (Theorem B). Specifically, if the flow category $\mathcal{C}$ is bounded below, it induces a bounded below chain complex in $\text{Ch}(\text{Mod}_{\mathcal{X}})$, which by Corollary 2.27 yields strong convergence. If we omit the bounded below condition on $\mathcal{C}$, we still obtain an exact couple that conditionally converges to the generalized homology of the realization, as detailed in Corollary 2.27.

resolve (what is the comp. cat. ?)

resolve: by Lurie we have that the S.S converge to the homotopy groups of the colim of the filtered object. We never justified why this colim is $|\mathcal{C}|$ move this ref after the thm?

7.1 Change of rings

Give a framed flow category $\mathcal{C}$, we can benefit from the fact that the changing the tangential structure of manifolds correspond to maps between spectra, to get a spectral sequence that calculates the bordism homology (represented by Thom spectra) of a manifold.

Let $\mathcal{X}, \mathcal{Y}$ be two tangential structures. Denote by $\mathbf{Flow}^{\mathcal{X}}$ (resp. $\mathbf{Flow}^{\mathcal{Y}}$) the collections of $\mathcal{X}$ (resp. $\mathcal{Y}$) structured flow categories.

A map $\mathcal{X} \rightarrow \mathcal{Y}$ induce a "forgetting" map F , from $\mathcal{X}$ manifolds to $\mathcal{Y}$ manifolds. The fact that Cartesian product commutes with F , induces a map:

$$(-)^{\mathcal{Y}} : \mathbf{Flow}^{\mathcal{X}} \rightarrow \mathbf{Flow}^{\mathcal{Y}}$$

On the other hand, the map $\mathcal{X} \rightarrow \mathcal{Y}$ induces a map of spectra $M\mathcal{X} \rightarrow M\mathcal{Y}$ and hence an extension of scalars of coherent chain complexes:

$$\otimes_{M\mathcal{X}} M\mathcal{Y} : \text{Ch}(\text{Mod}_{M\mathcal{X}}) \rightarrow \text{Ch}(\text{Mod}_{M\mathcal{Y}})$$

induces by post composition

$$\text{Ch} \xrightarrow{K} \text{Mod}_{M\mathcal{X}} \xrightarrow{\otimes_{M\mathcal{X}} M\mathcal{Y}} \text{Mod}_{M\mathcal{Y}}$$

Lemma 7.8. *The following diagram commutes:*

$$\begin{array}{ccc} \mathbf{Flow}^{\mathcal{X}} & \xrightarrow{K_{(-)}} & \text{Ch}(\text{Mod}_{M\mathcal{X}}) \\ (-)^{\mathcal{Y}} \downarrow & & \downarrow \otimes_{M\mathcal{X}} M\mathcal{Y} \\ \mathbf{Flow}^{\mathcal{Y}} & \xrightarrow{K_{(-)}} & \text{Ch}(\text{Mod}_{M\mathcal{Y}}) \end{array}$$

Re-write the proof

Proof. Let $\mathcal{C}$ be a $\mathcal{X}$ flow category. On the one hand we have $K_{\mathcal{C}^{\mathcal{Y}}}$, on the other $K_{\mathcal{C}} \otimes_{M\mathcal{X}} M\mathcal{Y}$. The object of the coherent chains are:

$$(K_{\mathcal{C}^{\mathcal{Y}}})_p = \bigoplus_{|z|=p} M\mathcal{Y} = \bigoplus_{|z|=p} M\mathcal{X} \otimes_{M\mathcal{X}} M\mathcal{Y} \simeq (K_{\mathcal{C}} \otimes_{M\mathcal{X}} M\mathcal{Y})_p.$$

We have that the maps of $K_{\mathcal{C}^{\mathcal{Y}}}$ are the collapse maps:

$$c_{M_{z,w}^{\mathcal{Y}}} : \Sigma_+^{\infty} \mathbb{R}_{0 \leq}^{J_{z,w}} \rightarrow M\mathcal{Y} \quad (40)$$

where $M^{\mathcal{Y}}$ is the manifold M with $\mathcal{Y}$ structure. The maps of $K_{\mathcal{C}} \otimes_{M\mathcal{X}} M\mathcal{Y}$ are given by collapsing and tensoring

$$c_{M_{z,w}} \otimes_{M\mathcal{X}} M\mathcal{Y} : \Sigma_+^{\infty} \mathbb{R}_{0 \leq}^{J_{z,w}} \rightarrow M\mathcal{X} \xrightarrow{(-) \otimes_{M\mathcal{X}} M\mathcal{Y}} M\mathcal{Y}. \quad (41)$$

The fact that the $\mathcal{Y}$ structure of $M_{z,w}$ is coming from $\mathcal{X}$ yields that the collapse map of $M_{z,w}^{\mathcal{Y}}$ factors through $\mathcal{X}$, the functoriality of the collapse map give the equivalence between Eq. (40) and Eq. (41). $\square$

First, we get that we can start with framed flow category and forget to other tangential structure, to get that:

Corollary 7.9. *Let $\mathcal{C}$ be a framed flow category and let $\mathcal{X}$ be a tangential structure. Then the spectral sequence of [Theorem A](#) implies to $\mathcal{C}^{\mathcal{X}}$ having:*

$$E_r \implies \pi_*(|\mathcal{C}^{\mathcal{X}}|) \simeq \pi_*(|\mathcal{C}| \otimes M\mathcal{X}) =: M\mathcal{X}_*(|\mathcal{C}|)$$

The first equivalence holds because the realization from Ch to $\widehat{\text{Fil}}$ is given by colimit, hence commutes with tensoring (as left adjoint).

Second, we can apply it to Morse function, to calculate the bordism homology of a manifold:

Corollary 7.10. *Let $f : M \rightarrow \mathbb{R}$ be a Smale Morse function and let $\mathcal{X}$ be a tangential structure, then the spectral sequence of the framed flow category $(\mathcal{C}_f)^{\mathcal{X}}$ converge to $M\mathcal{X}_*(M)$.*

Remark 7.11. Since the Morse index is inherently bounded below (by 0), the induced flow category $\mathcal{C}_f$ is bounded below. Thus, [Corollary 2.27](#) guarantees strong convergence to the bordism homology.

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